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RegTech Adoption and the Cost of Capital

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Abstract. This paper studies the cost of capital effect of a major regulatory technology, or RegTech, event: the staggered implementation of the Electronic Data Gathering, Analysis, and Retrieval (EDGAR) system of the Securities and Exchange Commission in the period from 1993 to 1996. This event represents a largely exogenous shock to corporate information dissemination technologies, resulting in a considerable reduction in information acquisition costs for investors. Using a difference-in-differences research design, we show that the cost of equity capital declines substantially after a firm switches from paper filing to mandatory electronic filing in EDGAR. The effect is stronger for small firms and firms with low analyst coverage and low institutional ownership. We identify three channels through which EDGAR affects a firm's cost of capital: the liquidity, risk-taking, and corporate governance channels. EDGAR implementation also improves a firm's investment efficiency significantly. We find evidence that the marginal value of a firm's capital investment and cash is higher during the post-EDGAR period.

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Keywords: RegTech • information acquisition cost • cost of capital • liquidity • corporate governance • EDGAR

1. Introduction

The issue of how the adoption of new technologies impacts financial markets and capital allocation is central to the financial economics literature. In particular, *technology-enabled innovation in financial regulations* (or, more succinctly, “RegTech,” short for regulatory technology) has greatly transformed the regulatory environment within which companies operate. Against this backdrop, this study examines how a recent RegTech event affects firms' costs of capital and investment efficiency.

We focus on the staggered implementation of the Electronic Data Gathering, Analysis, and Retrieval (EDGAR) system of the U.S. Securities and Exchange Commission (SEC) in the period from 1993 to 1996, which is arguably the first major RegTech event in recent times. This new regulation requires firms to file financial statements electronically via EDGAR rather than mailing paper copies directly to the SEC, as was done previously.¹ Earlier studies by Qi et al. (2000), Asthana and Balsam (2001), and Asthana et al. (2004) collectively show that 10-K filings on EDGAR contain new information and that not only does trading volume increase around filing dates but also

trades become more informative after EDGAR adoption. Recent studies further show that EDGAR implementation and the resultant reduction in the cost of accessing mandatory corporate filings lead to more private information production by corporate outsiders (Gao and Huang 2020) and that fund managers generate abnormal returns from processing information downloaded from the SEC's EDGAR servers (Chen et al. 2020). These findings are consistent with the argument that some investors are capable of processing public disclosures into private information (Blankespoor et al. 2020), as well as that the enhanced accessibility of public disclosures from EDGAR stimulates more private information production.

Prior theoretical research provides insights on the potential impact of EDGAR implementation on firms' costs of capital. The literature on limited stock market participation (e.g., Merton 1987, Basak and Cuoco 1998, and Shapiro 2002) posits that only a restricted set of investors participate in trading any particular stock because of the information cost investors must pay to become aware of a firm. Hence, a reduction in

information costs should broaden a firm's investor base and reduce its cost of equity (Merton 1987).

The literature on information acquisition in competitive markets (e.g., Grossman and Stiglitz 1980 and Verrecchia 1982) suggests that a reduction in information acquisition costs induces more investors to become informed, and because price aggregates informed traders' information and partially reveals it, in equilibrium, the stock price becomes more informative as a result of more informed trading. The increased information in the market can reduce the estimation risk associated with investors' assessments of the stock's future payoff, lowering its required rate of return (Barry and Brown 1985). Likewise, Lambert et al. (2007) also suggest that increased information about a firm reduces its assessed covariances with other firms' cash flows, moving its cost of equity closer to 0.

EDGAR implementation might also impact a firm's governance processes and therefore its cost of capital. First, enhanced stock price informativeness can strengthen the internal control mechanism by making managerial compensation more tightly linked to firm performance and therefore motivate managers to create value for the firm.² Second, timely and cheaply available corporate financial information reduces outside shareholders' costs of monitoring managers' actions. *Ceteris paribus*, enhanced managerial incentives and reduced external monitoring costs should increase investors' willingness to finance the firm and reduce its cost of capital.

However, there are also opposing views on how a reduction in information acquisition costs may affect a firm's cost of capital. In particular, Easley and O'Hara (2004) argue that private information represents an undiversifiable information risk for uninformed traders, for which, in equilibrium, they demand compensation. The increased private information production by some agents after EDGAR implementation can, therefore, give rise to greater information asymmetry and a higher cost of capital. Furthermore, Blankespoor et al. (2014) find an overall increase in abnormal bid-ask spreads and a decrease in abnormal trading volume (especially for small trades) around 10-K filings when firms started reporting in XBRL (eXtensible Business Reporting Language) format in 2009. Their results indicate that although the XBRL mandate was intended by the SEC to reduce investors' data collection and processing costs, it might have increased information asymmetry among some traders. Overall, whether and how EDGAR implementation affects a firm's cost of capital is an open empirical question.

We highlight that not all RegTech events are the same. The initial investment in technology and knowledge required to collect and analyze data reported in XBRL means that small investors are at a disadvantage. Indeed, the evidence from Blankespoor et al. (2014) is consistent with large investors possessing

better skills and having more resources for learning and processing XBRL taxonomy. By contrast, it is relatively straightforward to download filings from EDGAR for all investors alike, suggesting that small investors might actually experience a greater reduction in information acquisition costs than large investors, given the difficulty and high cost involved in accessing SEC filings prior to EDGAR. In light of the uniqueness of each RegTech event, it is useful to examine a setting other than XBRL to better understand how differences in the rules and implementation of RegTech might affect corporate outcomes.

The working assumption of our study is that the staggered implementation of the SEC's EDGAR system represents a largely exogenous shock to corporate information dissemination technologies, which plausibly leads to reduced information acquisition costs for investors. We provide evidence consistent with this assumption. Prior to the implementation of the EDGAR system, all publicly listed firms had to submit their filings to the SEC in paper format, and most investors could view these filings only in the SEC's public reference rooms located in Chicago, New York, and Washington, DC. Although some investors could obtain selected corporate disclosures from data vendors, the access was incomplete and highly costly. The SEC introduced the EDGAR system in February 1993 and mandated that all publicly listed firms begin filing their mandatory corporate disclosures (such as Forms 10-K and 10-Q) electronically through EDGAR in 10 discrete groups, with each group phased in on a different date. The first group started electronic filing in April 1993, and the last group started in May 1996.

Because the SEC randomly assigned firms to groups, the staggered inclusion of firms into the EDGAR system allows us to employ a difference-in-differences (DiD) approach to compare the cost of capital for a set of treated firms that had just switched to electronic EDGAR filing to a set of nonswitchers (i.e., the control firms) during the period.³ This research design has two advantages. First, it can disentangle the effect of a change in information costs on a firm's cost of capital from other confounding factors. For an omitted variable to bias our results, it has to affect the cost of capital of different groups of treated firms at different times in accordance with their EDGAR phase-in schedule. In other words, factors unrelated to the phase-in schedule will not confound our findings. Second, because the implementation of the EDGAR system represents a largely exogenous shock unrelated to any strategic actions taken by firms, it precludes the usual reverse causality concerns in empirical research on corporate outcomes.

We first examine the effect of the change to EDGAR filing on a firm's cost of equity capital at the quarterly frequency from Q2 1992 to Q2 1997. We find strong evidence that the transition from paper filing to electronic

filing in EDGAR reduces a firm's cost of equity. Relative to control firms that experience no change in their filing method during the period, the average new EDGAR filer experiences an economically and statistically significant decline of 39.5 basis points in the annual cost of equity capital, which is approximately 6.95% (6.42%) of its standard deviation (mean). The effect is even stronger for small firms and those with low pre-EDGAR analyst following and low institutional ownership. These results are robust to controlling for a host of firm characteristics and firm and year-quarter fixed effects.

We then investigate the channels through which the change to EDGAR filing affects a firm's cost of capital. The evidence points to three channels: liquidity, risk taking, and corporate governance. Our results show that a firm's monthly Amihud (2002) illiquidity measure declines by 5% (approximately 5.56% of its standard deviation) and effective bid-ask spread declines by 16.8 basis points (approximately 7.29% of its standard deviation) after EDGAR adoption. The results are broadly consistent with Merton (1987) in that reduced information costs lead to more trading and liquidity in a stock. The findings are also consistent with the argument that the increased accessibility of company information reduces information asymmetry and the adverse selection component in bid-ask spreads (Glosten and Milgrom 1985, Huang and Stoll 1997). In addition, we find that firms become less risky after the transition to EDGAR filing. For example, the expected default probability (based on the Merton distance to default model) declines by 0.08 percentage points on average—approximately 6.25% of its standard deviation—for EDGAR filers relative to their control firms. We also find evidence that managers engage in less earnings management after their mandatory EDGAR filing. This finding suggests that the enhanced accessibility of company information for investors creates a disciplinary effect on managers, pointing to a governance role of RegTech.

Last, we examine the effect of EDGAR implementation on a firm's investment efficiency. We employ the methodology suggested by Masulis et al. (2009) to compare the contribution of one extra dollar of capital expenditure and cash to firm value before and after a firm's transition to EDGAR filing. We find that a one-standard-deviation increase in the quarterly change in capital expenditure raises EDGAR filing firms' quarterly abnormal stock returns by approximately 1.5 percentage points relative to control firms. Similarly, the marginal value of cash also increases significantly after a firm switches to EDGAR filing.

We conduct several additional tests. We examine the parallel-trends assumption of our identification strategy by conducting two tests. In the first, we show that the pre-EDGAR time dummies are not significantly

related to the cost of capital. In the second, we employ a pseudo-EDGAR implementation schedule, set as two or four years prior to the actual schedule or, alternatively, as two years after the actual schedule. We find insignificant EDGAR effects in all three cases. Furthermore, our main research design follows a propensity score-matching (PSM) method to construct a sample of treated and control firms that share similar traits before EDGAR implementation and then employs a stacked DiD regression model (in which all cohorts of treated and control firms are aligned by event time and stacked together in one regression, following Cengiz et al. (2019)) to compare the outcome variables for the two groups of firms. Repeating the same analysis using a general regression-based DiD approach (following Armstrong et al. (2012) and Gao and Huang (2020)) or event-study DiD specifications (following Callaway and Sant'Anna (2021)) yields similar results.

The rest of this paper proceeds as follows. Section 2 surveys the related literature. Data and variable construction are described in Section 3. Section 4 documents the cost of the capital effect of EDGAR implementation, and Section 5 discusses the transmission channels. Section 6 examines the impact of EDGAR adoption on corporate investment efficiency. Robustness tests are in Section 7, and Section 8 concludes the paper.

2. Related Literature

Our paper contributes to the nascent literature examining the effect of RegTech on financial markets and capital allocation. Qi et al. (2000) and Asthana and Balsam (2001) are among the earliest studies showing that 10-K filings on EDGAR contain new information and investors trade more aggressively on news revealed by those filings during the post-EDGAR period. Asthana et al. (2004) further document a substantial increase in small trades after EDGAR adoption and show that these trades are informative. Gao and Huang (2020) document an increase in private information production by stock analysts and some individual investors post-EDGAR, resulting in an overall increase in stock price informativeness. Goldstein et al. (2022) find that the investment-to-price sensitivity declines significantly after a firm changes to EDGAR filing. Unlike our paper, these prior studies have not examined whether and how EDGAR adoption impacts a firm's cost of capital. In addition, we contribute to this strand of the literature by documenting a significant increase in the marginal value of a firm's capital investment and cash during the post-EDGAR period.⁴ Given the recent international rollouts of EDGAR-like systems (McClure et al. 2022), the findings from the current study are also instructive to countries that are considering implementing such systems or are in the process of doing so.

The SEC's mandate in 2009 that requires financial statements to be filed using XBRL represents another recent RegTech event. Dong et al. (2016) show that stock price synchronicity declined after XBRL adoption. Blankespoor et al. (2014) find evidence consistent with large investors (presumably institutional investors) benefiting more than small investors (presumably retail investors) from the XBRL mandate. Bhattacharya et al. (2018) show that among institutional investors, small traders benefit more than large traders. The existing literature, however, finds mixed evidence regarding the cost of capital effect of XBRL. For example, Blankespoor et al. (2014) document an overall *increase* in bid-ask spreads (around 10-K filings) for XBRL adopters relative to control firms, but Li et al. (2012) find the opposite result—they show an increase in stock liquidity and a *decrease* in the cost of capital after a firm starts XBRL reporting.⁵ Liu et al. (2014) and Chen et al. (2015) also document conflicting results for the XBRL effect in China.⁶ Given these mixed findings, it is useful to examine EDGAR implementation as another regulatory technology to better understand the market outcomes of changes in information costs.

We highlight that the information costs referenced throughout this paper are the same as disclosure processing costs (consisting of awareness, acquisition, and integration costs) discussed in Blankespoor et al. (2020). For small investors, all three types of disclosure processing costs are likely to decline after EDGAR adoption. For large investors, disclosure acquisition and integration costs may also decline, albeit to a lesser extent.⁷

Our work is also related to a large body of literature examining the determinants of the cost of equity. The classic capital asset pricing model states that the cost of equity depends solely on a firm's exposure to the market portfolio. Subsequent studies identify additional relevant risk factors, such as illiquidity risk (Amihud and Mendelson 1986), information risk (Easley and O'Hara 2004), and default risk (Chava and Purnanandam 2010). In addition, to the extent that agency costs reduce shareholder wealth, any internal or external control mechanisms that help mitigate agency problems should also reduce a firm's cost of equity (Shleifer and Vishny 1997). Our study adds to this strand of the literature by providing novel empirical evidence that EDGAR adoption—a RegTech event providing investors free online access to corporate filings—substantially reduces a firm's cost of equity via liquidity, risk taking, and corporate governance channels.

3. Data

3.1. The EDGAR System and Its Phase-in Schedule

As discussed in the introduction, prior to the introduction of the EDGAR system, investors could only

view mandatory corporate filings in the SEC's public reference rooms in either paper or microfiche format. Although a few data vendors offered access to *select* SEC filings of *select* firms to subscribers, the overall charge was prohibitively high to many investors.⁸

In 1983, the SEC embarked on an initiative to develop an electronic disclosure system. A pilot system started in May 1984 and concluded in July 1992. On February 23, 1993, the SEC issued rules to mandate that all publicly listed domestic firms begin filing their mandatory corporate disclosures electronically through EDGAR in 10 discrete groups, with each group phased in on a different date (SEC 1993). The first group of firms began electronic filing in April 1993, and the final group started in May 1996. (See Appendix A for the published schedule.)

Gao and Huang (2020) and SEC documents (SEC 1993 and SEC Release No. 33-6944 (1992)) indicate that the SEC assignment to the 10 EDGAR phase-in groups was random with the exception that voluntary filers (who participated in the pilot program to file some or all of their filings electronically) were assigned to the first group and that smaller firms were assigned to the final group to give them more time to adjust to the new filing method. We therefore exclude the first and the last group of firms from our empirical analysis.

3.2. Estimating the Cost of Equity Capital

We follow Hail and Leuz (2009) in using the simple average of estimates from four commonly used implied cost of capital (ICC) models (Claus and Thomas 2001, Gebhardt et al. 2001, Easton 2004, Ohlson and Juettner-Nauroth 2005) to proxy for the ex ante cost of equity capital. The details of our implementation of the four ICC models are summarized in Appendix B. We conduct several robustness checks to ensure that our results are not driven by measurement errors in the estimation of the ICC. First, our findings are robust to using the ICC estimate from any of the four models. Second, we also conduct analysis using the first principal component of the estimates obtained from these four models. Third, instead of assigning equal weight to the four ICC estimates, we weight them by the fit (i.e., regression R^2) of each individual ICC model. Fourth, we estimate a weighted ordinary least squares (OLS) regression using the accuracy of analysts' forecasts as the weight, following Hail and Leuz (2009).

3.3. Variable Construction and Summary Statistics

We obtain accounting data from Standard & Poor's quarterly Compustat database, market data from the Center for Research in Security Prices (CRSP), and analysts' forecast data from the Institutional Brokers Estimate System (I/B/E/S) database. To estimate the

ICC, we keep firm-quarters with nonmissing values on the book value of equity, numbers of shares outstanding, earnings, dividends, and analysts' forecasts on one-year- and two-year-ahead earnings per share and long-term earnings growth. Firms are required to have at least one quarter of data both before and after EDGAR adoption to be included in our sample. We use the accounting data from the most recent fiscal year, ending at least three months before the ICC calculation, to ensure that the accounting information is publicly available and incorporated into stock prices at the time we estimate the ICC. We constrain the ICC estimates to be positive and treat negative estimates as missing values. The risk-free rate is proxied by the annualized one-month T-bill rate. We estimate the ICC at the end of each quarter based on the most recent analysts' forecasts for the sample period from Q2 1992 to Q2 1997 (i.e., from one year before the start date of the first group of EDGAR filers to one year after the start date of the final group). Our main sample includes 1,414 propensity score-matched pairs of treated and control firms, with a total of 15,500 firm-quarter observations.

The use of the I/B/E/S data imposes some limitations on our study. First, because we require firms

to have analyst forecast data from I/B/E/S to be included in our sample, our findings apply to only those firms covered by I/B/E/S analysts. We caution not to generalize our evidence to smaller firms without any analyst following. Second, analysts may not always update their forecasts in a timely manner. If staleness in analyst forecasts represents some random errors, it will bias against us finding any significant EDGAR effect.

We construct three monthly measures of stock liquidity: the Amihud illiquidity measure, relative quoted spread, and relative effective spread, following Amihud (2002), Chordia et al. (2001), and Lesmond et al. (1999). The quarterly expected default probability is estimated based on the Merton distance to the default model, following the procedure described in Bharath and Shumway (2008). We also construct three measures of absolute discretionary accruals estimated based on the modified Jones (1991), Dechow and Dichev (2002), and McNichols (2002) models.

Table 1 shows that the COC (the average ICC in excess of the risk-free rate) has a mean (median) of 6.15% (5%) per annum, with the difference between the 25th and 75th percentiles at a magnitude of approximately

Table 1. Summary Statistics

Variable	No.	Mean	Std. dev.	P25	Median	P75	Variable	No.	Mean	Std. dev.	P25	Median	P75
Panel A: Quarterly frequency													
COC (%)	15,500	6.151	5.686	2.636	5.001	7.890	R&D	21,365	0.010	0.023	0.000	0.000	0.010
Treat	19,762	0.519	0.500	0.000	1.000	1.000	StdCF	21,365	0.038	0.038	0.017	0.028	0.047
Post	19,762	0.509	0.500	0.000	1.000	1.000	StdSales	21,365	0.058	0.050	0.023	0.044	0.077
logSize	19,762	5.704	2.103	4.124	5.588	7.166	Tangibility	21,365	0.302	0.230	0.118	0.243	0.440
BM	19,762	0.616	0.393	0.343	0.544	0.783	Fscore	21,365	0.831	0.515	0.413	0.718	1.142
Leverage	19,762	0.234	0.166	0.098	0.212	0.344	HighF_20p	21,365	0.209	0.406	0.000	0.000	0.000
Ret	19,762	0.024	0.188	−0.077	0.022	0.123	IO	15,047	0.269	0.238	0.058	0.208	0.444
FBias	15,500	−0.014	0.060	−0.008	−0.000	0.002	Sales growth	19,471	0.137	1.908	−0.038	0.038	0.129
R ^{PC} (%)	15,500	12.449	11.318	5.489	10.257	15.710	r − R ^{IND}	17,918	−0.012	0.199	−0.129	−0.024	0.083
R ^{Model fit} (%)	15,500	6.078	5.258	2.859	5.063	7.711	r − R ^{Size, BM}	18,088	−0.000	0.192	−0.115	−0.014	0.090
R ^{FE} (%)	15,500	6.001	5.717	2.549	4.817	7.653	ΔCAPX	16,902	0.001	0.018	−0.004	0.000	0.005
R ^{ExtFin} (%)	15,500	5.841	5.640	2.390	4.654	7.482	CAPX	16,902	0.018	0.023	0.005	0.012	0.022
R ^{Unaffiliated} (%)	15,500	5.962	5.823	2.446	4.725	7.554	ΔCash	18,088	0.003	0.068	−0.014	0.000	0.013
R ^{PSS} (%)	13,152	7.861	8.908	1.805	5.574	10.667	Cash	18,088	0.134	0.173	0.017	0.060	0.186
R ^{PEG} (%)	8,939	7.257	5.592	3.596	6.630	9.762	Net financing	18,088	0.013	0.069	−0.006	0.000	0.006
Stock return volatility (StdRet)	19,011	0.416	0.222	0.249	0.364	0.529	ΔEarnings	18,088	0.001	0.028	−0.006	0.001	0.007
Expected default probability	19,762	0.069	0.128	0.000	0.001	0.076	ΔNetAsset	18,088	0.030	0.097	−0.012	0.015	0.055
Accruals ^{Mod Jones}	13,215	3.533	4.572	0.972	2.151	4.370	ΔR&D	18,088	0.000	0.009	0.000	0.000	0.000
Accruals ^{DD}	18,159	2.051	2.600	0.560	1.312	2.739	ΔInterest	18,088	0.000	0.001	−0.000	0.000	0.000
Accruals ^{MC}	14,497	2.357	3.133	0.539	1.348	2.915	ΔDividend	18,088	0.000	0.002	0.000	0.000	0.000
Age	21,365	20.093	12.684	10.000	16.000	29.000	Interim	15,500	0.093	0.290	0.000	0.000	0.000
ROA	21,365	0.004	0.047	0.002	0.012	0.023							
Panel B: Monthly frequency													
logAmihud	63,285	0.585	0.954	0.011	0.101	0.744	Leverage	63,285	0.214	0.185	0.045	0.186	0.337
Effective bid-ask spread (%)	37,145	2.236	2.306	0.613	1.407	3.006	Ret	63,285	0.174	1.480	−0.667	0.000	0.857
Quoted bid-ask spread (%)	37,145	3.127	3.182	0.862	1.938	4.239	StdRet	63,285	0.523	0.357	0.273	0.428	0.661
logSize	63,285	5.081	2.003	3.566	5.013	6.437	Prc	63,285	18.306	16.030	6.250	13.750	25.688
BM	63,285	0.566	0.392	0.291	0.481	0.743							

Notes. This table reports the summary statistics of the variables used in our main analyses with propensity score-matched samples. Panel A reports the statistics at the quarterly frequency, and panel B does so at the monthly frequency. Detailed variable definitions are provided in Appendix C.

5.25%. These estimates are broadly in line with those reported in the literature (e.g., Hail and Leuz 2009, Hou et al. 2012, and Houston et al. 2018). The Amihud illiquidity measure is small and highly right-skewed, so we multiply it by 10^6 and then apply a natural log transformation. The transformed measure, $\log Amihud$, has a mean (median) of 0.585 (0.101) and a standard deviation of 0.954. Table 1 also shows that the average expected default probability and earnings management measures vary substantially across firms. Detailed definitions of all variables are provided in Appendix C.

4. Cost of Capital Effect of EDGAR Implementation

4.1. Baseline Results: Using the Full Sample

In this subsection, we employ a general DiD test methodology with the following specification:

$$COC_{i,t} = \alpha_i + \beta_t + \theta_1 PostEDGAR_{i,t} + \varepsilon_{i,t}, \quad (1)$$

where i indexes firms and t indexes year-quarters, α_i denotes firm fixed effects, and β_t denotes year-quarter fixed effects. The dependent variable $COC_{i,t}$ is the average annualized cost of equity (estimated from the

four ICC models) in excess of the risk-free rate for firm i in year-quarter t . Dummy variable $PostEDGAR_{i,t}$ that takes value 1 if firm i is subject to mandatory EDGAR filing by year-quarter t and 0 otherwise. The residual term is denoted by $\varepsilon_{i,t}$. Note that our regression specification in Equation (1) is identical to equation (4.1) in Athey and Imbens (2022). In theorem 1 of their paper, they show that under certain assumptions, the expected value of the OLS estimator $\hat{\theta}_1$ equals the population treatment effect of the policy (i.e., the population treatment effect of EDGAR adoption in our case).⁹

We estimate Equation (1) using the OLS approach, with standard errors clustered at the firm level to allow for unspecified correlation in the error terms over time. Table 2, column (1) reports the result. The estimate of $PostEDGAR_{i,t}$ is -0.313 , suggesting that the transition to EDGAR filing reduces a firm's cost of equity substantially at a magnitude of 31.3 basis points, approximately 5.11% (5.09%) of the standard deviation (mean) of the variable.

To further increase the precision of the regression estimates, we expand Equation (1) to include a set of firm-level controls, $X_{i,t-1}$. In addition, progress in information technology during our sample period might

Table 2. The Cost of Capital Effect of EDGAR Adoption (Full Sample)

Independent variable	COC (1)	COC (2)	COC (3)	R^{PC} (4)	$R^{Model\ fit}$ (5)	$R^{Accuracy}$ (6)
<i>PostEDGAR</i>	−0.313*** (−2.93)	−0.318*** (−2.99)	−0.333*** (−3.19)	−0.681*** (−3.35)	−0.319*** (−3.09)	−0.325*** (−3.53)
<i>logSize</i>			0.217 (1.29)	0.393 (1.24)	0.236 (1.39)	0.404*** (2.89)
<i>BM</i>			1.588*** (8.16)	3.458*** (9.15)	1.476*** (7.40)	1.288*** (7.61)
<i>Leverage</i>			3.211*** (6.53)	6.134*** (6.55)	3.059*** (6.19)	2.673*** (6.64)
<i>Ret</i>			−0.310*** (−10.79)	−0.566*** (−9.88)	−0.310*** (−10.85)	−0.311*** (−12.84)
<i>StdRet</i>			2.579*** (8.14)	5.162*** (8.56)	2.567*** (7.82)	2.480*** (9.22)
<i>FBias</i>			−1.752 (−1.46)	1.446 (0.69)	−2.662** (−1.98)	−2.291** (−2.08)
Industry time trend	N	Y	Y	Y	Y	Y
Year-quarter FE	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
Obs.	40,016	40,016	40,016	40,016	40,016	39,988
Adj. R^2	0.789	0.791	0.798	0.761	0.807	0.819

Notes. This table examines the effect of EDGAR adoption on a firm's cost of equity capital. We estimate the effect using the full set of panel data in our sample, which consists of 40,016 firm-quarter observations from Q2 1992 to Q2 1997. In columns (1)–(3), the dependent variable $COC_{i,t}$ is the cost of equity capital (proxied by the average estimate from the four ICC models) in excess of the risk-free rate (proxied by the annualized one-month T-bill rate) for firm i in year-quarter t . The main variable of interest is $PostEDGAR_{i,t}$, which is a dummy variable that takes value 1 if firm i is subject to mandatory EDGAR filing by year-quarter t and 0 otherwise. Because a firm's inclusion in the EDGAR system always starts in the middle of a quarter, we exclude that quarter from our main analysis. In column (4), the dependent variable is the first principal component of the four ICC estimates (labeled $R_{i,t}^{PC}$). In column (5), we use as the dependent variable a weighted average cost of equity capital in excess of the risk-free rate (labeled $R_{i,t}^{Model\ fit}$). Columns (1)–(5) are estimated using OLS regressions. In column (6), we repeat the estimation of column (3) but use a weighted OLS technique. To differentiate this weighted OLS specification from the baseline OLS regression reported in column (3), we rename the dependent variable in this specification $R_{i,t}^{Accuracy}$. All of the cost of capital proxies are measured in percentages. Robust t -statistics are reported in parentheses, calculated using standard errors clustered at the firm level. A detailed description of the four ICC models is in Appendix B. Detailed variable definitions are provided in Appendix C. FE, fixed effects.

** and *** denote statistical significance at the 5% and 1% levels, respectively.

have led to heterogeneous declines in the cost of capital across different industries. We thus follow Moser and Voena (2012) to include a separate time trend for each industry, as follows:

$$COC_{i,t} = \alpha_i + \beta_t + \theta_1 PostEDGAR_{i,t} + \gamma' X_{i,t-1} + \varphi I_c \times t + \varepsilon_{i,t}, \quad (2)$$

where I_c is an indicator variable for industry c , with the industry assignment being based on the Fama–French industry classification with 48 industries. We then multiply the indicator variable by t (with a slight abuse of notation, the value of t here is set to equal 1 for the first quarter, 2 for the second quarter, etc.) to allow for industry-specific time trends. The term $X_{i,t-1}$ is a vector of firm characteristics measured in quarter $t - 1$, including the natural logarithm of total book assets ($\log Size_{i,t-1}$), book value to market value of equity ($BM_{i,t-1}$), leverage ratio ($Leverage_{i,t-1}$), the previous quarter's stock returns ($Ret_{i,t-1}$), standard deviation of the previous quarter's daily returns ($StdRet_{i,t-1}$), and analyst forecast bias ($FBias_{i,t-1}$), following Hail and Leuz (2009) and Chava and Purnanandam (2010).¹⁰ All control variables are winsorized at the 1% and 99% levels to minimize the effect of outliers.

Columns (2) and (3) of Table 2 report the estimation result of Equation (2). The estimate of $PostEDGAR_{i,t}$ is -0.318 (t -stat = -2.99) when control variables $X_{i,t-1}$ are not included, and it increases slightly to -0.333 (t -stat = -3.14) when control variables are included. In column (4), we replace $COC_{i,t}$ with the first principal component of the four ICC estimates (labeled $R_{i,t}^{PC}$) and then reestimate Equation (2). The coefficient of $PostEDGAR_{i,t}$, -0.681 , is again highly statistically significant. In column (5), we calculate a weighted-average cost of equity capital (labeled $R_{i,t}^{Model\ fit}$), which uses the regression R^2 of each individual ICC model as the weight. In column (6), we reestimate Equation (2) using a weighted OLS technique, with the inverse of the absolute one-year-ahead analyst forecast error (scaled by lagged total assets) as the weight. To differentiate this weighted OLS specification from the baseline OLS regression reported in column (3), we rename the dependent variable in this specification $R_{i,t}^{Accuracy}$. Our result remains robust to these two alternative regression specifications.

4.2. Baseline Results: Using the Propensity Score–Matched Sample

If firm assignment to groups is completely random, the OLS estimator $\hat{\theta}_1$ in Equation (1) for the EDGAR treatment effect is unbiased and consistent. In the scenario in which firm assignment is random only after conditioning on some observable firm characteristics, one can obtain an unbiased and consistent estimate of the EDGAR treatment effect using the regression

specification in Equation (2) as long as the conditioning variables are included in $X_{i,t-1}$ (Roberts and Whited 2013). An alternative way of addressing this “selection on observables” issue is to employ a PSM method. Angrist and Pischke (2008) suggest that because these two methods rely on the same core assumption (i.e., unconfoundedness and overlap) for causal inference, they are unlikely to have any material empirical differences.

In this subsection, we seek to establish our baseline results using the PSM approach. We first identify a matching firm for each treated firm, following a similar method employed by Gao and Huang (2020). Specifically, for each quarter, when a group of firms switches to EDGAR filing, we create a cohort consisting of treated firms (i.e., firms that change from paper filing to electronic filing during the quarter) and control firms (i.e., EDGAR filers or nonfilers that have not changed their filing method from four quarters prior to this event quarter to four quarters afterward). We then pool all eight cohorts of treated and control firms into a panel and estimate a logistic regression to predict the likelihood that a firm becomes treated using a list of key firm characteristics and cohort and industry fixed effects. Then, we use the predicted probabilities (namely, the estimated propensity scores) from this logit estimation to perform one-to-one nearest-neighbor matching with replacement. This process generates 1,414 matched pairs of treated and control firms.

Panel A of Table 3 reports the balance test result on treated firms and their matched control firms. Columns (1) and (2) report the mean of each firm characteristic for treated firms and control firms, respectively. The mean difference between the two groups is reported in column (3), and the p -value for the test of the equality of their means is reported in column (4). The corresponding descriptive statistics and equality test results using median values are reported in columns (5)–(8). Both sets of tests indicate that treated firms and control firms are statistically indistinguishable along the 12 matching characteristics.

Next, we align the eight cohorts of matching treated and control firms by event time and then estimate the following regression model, using data from four quarters prior to the event quarter to four quarters afterward, with the event quarter excluded:

$$COC_{i,t} = \alpha_i + \beta_t + \delta_g + \delta_1 Treat_i \times Post_{i,t} + \delta_2 Treat_i + \gamma' X_{i,t-1} + \varphi I_c \times t + \varepsilon_{i,t}, \quad (3)$$

where $Treat_i$ is a dummy variable for treated firms; $Post_{i,t}$ is a post-EDGAR dummy; and $COC_{i,t}$, $X_{i,t-1}$, and $I_c \times t$ are the same as those in Equation (2). Let α_i , β_t , and δ_g denote firm, year-quarter, and cohort fixed effects, respectively. (Without loss of clarity, we omit the cohort subscript g from the variables $Treat_i$ and $Post_{i,t}$ for simplicity.) Panel B of Table 3 reports the

Table 3. The Cost of Capital Effect of EDGAR Adoption (Propensity Score–Matched Sample)

Panel A: Balance test								
Firm characteristic	Mean				Median			
	Treat (1)	Control (2)	Difference (3)	<i>p</i> -value (4)	Treat (5)	Control (6)	Difference (7)	<i>p</i> -value (8)
<i>logSize</i>	5.965	5.922	0.043	0.544	5.750	5.851	−0.101	0.546
<i>BM</i>	0.578	0.577	0.001	0.946	0.521	0.511	0.010	0.972
<i>Leverage</i>	0.160	0.156	0.004	0.491	0.113	0.099	0.014	0.635
<i>Ret</i>	0.098	0.105	−0.007	0.783	0.207	0.191	0.016	0.938
<i>StdRet</i>	0.094	0.092	0.002	0.218	0.083	0.082	0.001	0.162
<i>FBias</i>	−0.019	−0.017	−0.002	0.445	0.000	0.000	0.000	0.215
<i>ROA</i>	0.009	0.010	−0.001	0.336	0.010	0.011	−0.001	0.165
<i>Tangibility</i>	0.285	0.273	0.012	0.210	0.216	0.191	0.025	0.175
<i>Sales growth</i>	0.054	0.047	0.007	0.492	0.039	0.034	0.005	0.620
<i>CAPX</i>	0.015	0.015	0.000	0.271	0.010	0.009	0.001	0.181
<i>R&D</i>	0.008	0.007	0.001	0.267	0.000	0.000	0.000	0.766
<i>IO</i>	0.279	0.281	−0.002	0.800	0.257	0.238	0.019	0.885
Panel B: The propensity score–matched results								
Independent variable	COC (1)	COC (2)	COC (3)	<i>R</i> ^{PC} (4)	<i>R</i> ^{Model fit} (5)	<i>R</i> ^{Accuracy} (6)		
<i>Treat</i> × <i>Post</i>	−0.308** (−2.47)	−0.307** (−2.46)	−0.395*** (−3.19)	−0.967*** (−3.64)	−0.355*** (−3.20)	−0.325*** (−3.06)		
<i>Treat</i>	0.019 (0.14)	0.011 (0.08)	0.087 (0.66)	0.352 (1.30)	0.067 (0.57)	0.086 (0.73)		
<i>logSize</i>			−0.350 (−1.11)	−0.399 (−0.61)	−0.095 (−0.35)	−0.080 (−0.31)		
<i>BM</i>			1.417*** (3.96)	2.763*** (3.72)	1.157*** (3.93)	1.083*** (3.34)		
<i>Leverage</i>			4.035*** (3.62)	7.888*** (3.64)	3.144*** (3.34)	2.644*** (3.14)		
<i>Ret</i>			−0.277*** (−6.03)	−0.509*** (−4.53)	−0.255*** (−5.67)	−0.295*** (−7.31)		
<i>StdRet</i>			2.633*** (4.58)	6.231*** (5.21)	2.346*** (4.81)	2.327*** (5.06)		
<i>FBias</i>			−1.794 (−0.80)	2.782 (0.68)	−0.502 (−0.31)	−0.814 (−0.41)		
Industry time trend	N	Y	Y	Y	Y	Y		
Year-quarter FE	Y	Y	Y	Y	Y	Y		
Cohort FE	Y	Y	Y	Y	Y	Y		
Firm FE	Y	Y	Y	Y	Y	Y		
Obs.	15,500	15,500	15,500	15,500	15,500	15,500		
Adj. <i>R</i> ²	0.838	0.84	0.845	0.816	0.855	0.861		

Notes. This table examines the effect of EDGAR adoption on a firm's cost of equity capital using the propensity score–matched sample. The sample consists of 1,414 pairs of treated firms and control firms. Panel A reports the balance test results for the matching characteristics used in the logistic regression. Columns (1) and (2) report the mean of each firm characteristic for treated firms and control firms, respectively. The mean difference between the two groups is reported in column (3), and the *p*-value for the test of equality of their means is reported in column (4). The corresponding statistics and equality test results using median values are reported in columns (5)–(8). In panel B, we align the eight cohorts of matching treated and control firms by event time and then estimate stacked DiD regression models. The main variable of interest is the interaction term $Treat_i \times Post_{i,t}$, where $Treat_i$ is a dummy variable for treated firms and $Post_{i,t}$ is a post-EDGAR dummy. All the dependent and control variables are defined in the same way as those in Table 2. All regression models use data from four quarters prior to the event quarter to four quarters afterward, with the event quarter excluded. Robust *t*-statistics are reported in parentheses, calculated using standard errors clustered at the firm level. Detailed variable definitions are provided in Appendix C. FE, fixed effects.

** and *** denote statistical significance at the 5% and 1% levels, respectively.

results. The dependent variable in columns (1)–(3) is the implied cost of capital $COC_{i,t}$. As shown in column (3), with the full set of control variables, the estimate of the interaction term $Treat \times Post$ is −0.395, which is statistically significant at the 1% level and quantitatively close to the estimate of −0.333 reported in Table

2, column (3). As in the previous subsection, we also test the robustness of our baseline result using alternative measures of the cost of capital as well as a weighted OLS technique, with the results reported in columns (4)–(6). The estimates of the interaction term, $Treat \times Post$, are −0.967 (*t*-stat = −3.64), 0.355 (*t*-stat =

−3.2), and −0.325 (t -stat = −3.06) in columns (4), (5), and (6), respectively, all of which are highly significant at the 1% level.

In summary, both the general DiD approach (employed in the previous subsection) and the PSM method (employed in this subsection) provide qualitatively similar results: firms experience a substantial decline in their costs of equity capital after switching to online EDGAR filing. For brevity, we report only the results using the PSM method for the rest of our analyses.

4.3. Heterogeneous Effects Across Firms

Firms with less media and analyst coverage typically are associated with higher information costs for investors. Such firms are more opaque, and they tend to be smaller and have relatively more retail investors. The enhanced accessibility of corporate filings during the post-EDGAR period should be particularly beneficial for such firms. To test this hypothesis, we split our sample into two based on the median value of three different firm characteristics: firm size, institutional ownership (IO), and analyst coverage, measured prior to EDGAR implementation.

Table 4 reports the estimation results of Equation (3) separately for small, large, low IO , high IO , low analyst coverage, and high analyst coverage firms. We find consistent evidence that the EDGAR effect is stronger for small firms and those with low institutional ownership and low analyst coverage. The results show that small, low IO , and low analyst coverage firms, on average, experience a decline of 92.3, 66.2, and 59.4 basis points, respectively, in the cost of equity capital after EDGAR adoption, representing approximately 9%–13% of their respective mean. By contrast, large, high IO , and high analyst coverage firms show a much smaller decline, approximately 0%–4% of their respective mean. Overall, the evidence suggests that the EDGAR effect is stronger for firms with a less transparent pre-EDGAR information environment.

5. Channels Through Which EDGAR Filing Affects Financing Costs

5.1. Stock Liquidity

The investor recognition hypothesis advanced by Merton (1987) implies that a reduction in information costs should increase investor participation in the stock

Table 4. Subsample Analysis

Independent variable	Firm size		Institutional ownership		Analyst coverage	
	Small (1)	Large (2)	Low (3)	High (4)	Low (5)	High (6)
<i>Treat</i> × <i>Post</i>	−0.923*** (−4.07)	−0.001 (−0.00)	−0.662*** (−3.10)	−0.102 (−0.55)	−0.594*** (−2.93)	−0.223 (−1.36)
<i>Treat</i>	−0.187 (−0.68)	0.013 (0.10)	0.182 (0.82)	−0.035 (−0.19)	0.121 (0.54)	0.156 (1.10)
<i>logSize</i>	−0.672 (−1.51)	0.113 (0.25)	−0.597 (−0.91)	0.306 (1.00)	−0.866* (−1.71)	0.096 (0.24)
<i>BM</i>	2.189*** (4.39)	0.458 (0.91)	1.985*** (3.32)	1.203** (2.42)	1.727*** (3.27)	1.201*** (2.75)
<i>Leverage</i>	4.240** (2.36)	3.909*** (3.58)	4.589** (2.43)	1.109 (1.32)	5.274** (2.55)	2.693*** (2.88)
<i>Ret</i>	−0.341*** (−4.80)	−0.195*** (−3.59)	−0.251*** (−3.14)	−0.275*** (−4.55)	−0.351*** (−4.69)	−0.203*** (−3.68)
<i>StdRet</i>	2.735*** (3.66)	1.743** (2.27)	1.859** (2.49)	2.295*** (2.77)	2.534*** (3.24)	2.932*** (3.55)
<i>FBias</i>	2.885 (1.34)	−9.727** (−2.09)	−1.714 (−0.51)	6.432** (2.49)	0.092 (0.04)	−5.604 (−0.97)
Industry time trend	Y	Y	Y	Y	Y	Y
Year-quarter FE	Y	Y	Y	Y	Y	Y
Cohort FE	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
Obs.	6,554	8,896	5,487	6,360	6,754	7,995
Adj. R^2	0.813	0.880	0.722	0.809	0.801	0.901
χ^2 test	$\chi^2(1) = 10.68; p = 0.001$		$\chi^2(1) = 3.95; p = 0.05$		$\chi^2(1) = 2.06; p = 0.16$	

Notes. This table examines the heterogeneous effect of EDGAR implementation on different groups of firms: small, large, low IO , high IO , low analyst coverage, and high analyst coverage firms. The analysis uses the propensity score matched sample. The dependent variable is the cost of equity capital, $COC_{i,t}$, and the independent variables are the same as those in Table 3, column (3). The last two rows report the χ^2 statistic $\chi^2(1)$ and p -value for the null hypothesis of equal coefficients of the interaction term, *Treat* × *Post*, between small firms and large firms (or between firms with low and high institutional ownership or between firms with low and high analyst coverage). Robust t -statistics are reported in parentheses and are calculated using standard errors clustered at the firm level.

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

market. Consistent with this argument, Gao and Huang (2020) find an increase in retail trading volume following a firm's transition to EDGAR filing. More investors trading a stock can increase its liquidity. We therefore conjecture that stock liquidity might be a channel through which EDGAR impacts a firm's cost of capital.

We investigate this hypothesis using a stacked regression model similar to Equation (3). Because stock liquidity measures are available for a large set of firms and at a higher frequency, we have a total of 63,285 firm-month observations for treated and control firms combined.¹¹ Our empirical specification is as follows:

$$Iliquidity_{i,m} = \alpha_i + \beta_m + \vartheta_g + \delta_1 Treat_i \times Post_{i,m} + \delta_2 Treat_i + \gamma X_{i,m-1} + \varphi I_c \times m + \varepsilon_{i,m}, \quad (4)$$

where i indexes firms and m indexes year-months. The dependent variable, $Iliquidity_{i,m}$, represents one of the three stock illiquidity measures: the logarithmic Amihud illiquidity ratio ($\log Amihud$), relative effective bid-ask spreads, and relative quoted bid-ask spreads. The measures are constructed based on Amihud (2002), Chordia et al. (2001), and Lesmond et al. (1999). The variable $Treat_i$ is a dummy variable for treated firms; $Post_{i,m}$ is a post-EDGAR dummy that takes value 1 if

firm i is subject to mandatory EDGAR filing by year-month m and 0 otherwise. (Again, without loss of clarity, we omit the cohort subscript g from the variables $Treat_i$ and $Post_{i,m}$ for simplicity.) The terms α_i , β_m , and ϑ_g are firm, year-month, and cohort fixed effects, respectively, and I_c is an indicator variable for industry c . The indicator variable is then multiplied by m to allow for industry-specific time trends; again, with a slight abuse of notation, the value of m is set to equal 1 for the first month, 2 for the second month, and so on. For firm-level controls, $X_{i,m-1}$, we include the logarithmic total assets ($\log Size_{i,m-1}$), book value to market value of equity ($BM_{i,m-1}$), leverage ratio ($Leverage_{i,m-1}$), the previous month's stock returns ($Ret_{i,m-1}$), standard deviation of the previous month's daily returns ($StdRet_{i,m-1}$), and stock price ($Prc_{i,m-1}$) at the end of the month.¹²

Table 5 shows that both the high-frequency measures of stock illiquidity (i.e., relative effective and quoted spreads) and the low-frequency measure (i.e., the Amihud illiquidity ratio) decrease significantly after EDGAR adoption. The estimate in column (1) shows that the monthly Amihud illiquidity ratio drops by 5.2% (approximately 5.45% of its standard deviation

Table 5. EDGAR Filing and Stock Liquidity

Independent variable	$\log Amihud$ (1)	Effective bid-ask spread (2)	Quoted bid-ask spread (3)
$Treat \times Post$	-0.052*** (-3.46)	-0.168*** (-3.68)	-0.215*** (-3.52)
$Treat$	-0.041** (-2.42)	0.008 (0.19)	0.026 (0.45)
$\log Size$	-0.324*** (-12.89)	-0.602*** (-8.04)	-0.861*** (-8.99)
BM	0.586*** (16.05)	1.157*** (10.82)	1.567*** (11.64)
$Leverage$	0.520*** (7.14)	1.369*** (6.94)	1.907*** (7.12)
Ret	-0.052*** (-22.61)	-0.104*** (-18.87)	-0.156*** (-23.09)
$StdRet$	0.377*** (17.34)	0.987*** (15.36)	1.370*** (17.64)
Prc	0.002*** (3.68)	-0.007*** (-4.57)	-0.011*** (-4.90)
Industry time trend	Y	Y	Y
Year-month FE	Y	Y	Y
Cohort FE	Y	Y	Y
Firm FE	Y	Y	Y
Obs.	63,285	37,145	37,145
Adj. R^2	0.823	0.859	0.904

Notes. This table examines the effect of EDGAR implementation on stock liquidity. The analysis uses the propensity score-matched sample. The dependent variables are the natural logarithm of the monthly Amihud illiquidity ratio ($\log Amihud$), the relative effective bid-ask spread, and the relative quoted bid-ask spread in columns (1), (2), and (3), respectively. The variable of interest is the interaction term $Treat_i \times Post_{i,m}$, where $Treat_i$ is a dummy variable for treated firms and $Post_{i,m}$ is a post-EDGAR dummy that takes value 1 if firm i is subject to mandatory EDGAR filing by year-month m and 0 otherwise. The firm-level controls include the logarithmic total assets ($\log Size_{i,m-1}$), book value to market value of equity ($BM_{i,m-1}$), leverage ratio ($Leverage_{i,m-1}$), previous month's stock returns ($Ret_{i,m-1}$), standard deviation of the previous month's daily returns ($StdRet_{i,m-1}$), and stock price at the end of the month ($Prc_{i,m-1}$). Robust t -statistics are reported in parentheses and calculated using standard errors clustered at the firm level. FE, fixed effects.

** and *** denote statistical significance at the 5% and 1% levels, respectively.

and 8.89% of its mean) after firms switch to EDGAR filing. Columns (2) and (3) show a similar pattern for the other two measures of stock illiquidity, with the relative effective spreads dropping by 16.8 basis points (approximately 7.29% of the standard deviation and 7.51% of the mean) and relative quoted spreads by 21.5 basis points (approximately 6.76% of the standard deviation and 6.88% of the mean) for EDGAR filers. The overall results provide strong support for the stock liquidity channel of the EDGAR effect.

5.2. Corporate Risk Taking

Firm risk might decline after EDGAR implementation for two reasons. First, enhanced accessibility of company information during the post-EDGAR period might lead to more efficient external monitoring, which, in turn, improves managerial incentives and discourages excessive risk taking. Second, the increased private information production and overall stock price informativeness post-EDGAR might help firm managers make better investment decisions without taking undue risks. We therefore conjecture that a reduction in firm risk might be another channel through which EDGAR impacts a firm's cost of capital.

To investigate this channel, we consider two firm risk measures: stock return volatility and the expected default probability. Stock return volatility is estimated by the (annualized) standard deviation of daily stock returns in the quarter (French et al. 1987). We estimate the expected default probability based on the Merton distance to default model, following the procedure described in Bharath and Shumway (2008). We then estimate the following regression model for treated firms and their matched control firms:

$$\begin{aligned} \text{FirmRisk}_{i,t} = & \alpha_i + \beta_t + \vartheta_g + \delta_1 \text{Treat}_i \\ & \times \text{Post}_{i,t} + \delta_2 \text{Treat}_i + \gamma X_{i,t-1} + \varphi I_c \\ & \times t + \varepsilon_{i,t}, \end{aligned} \quad (5)$$

where i indexes firms and t indexes year-quarters. The dependent variable, $\text{FirmRisk}_{i,t}$, represents one of the two aforementioned proxies for firm risk. The terms α_i , β_t , ϑ_g , $I_c \times t$, Treat_i , and $\text{Post}_{i,t}$ are the same as those in Equation (3). The set of control variables $X_{i,t-1}$ includes $\log\text{Size}_{i,t-1}$, $BM_{i,t-1}$, $\text{Leverage}_{i,t-1}$, and $\text{Ret}_{i,t-1}$.

The results, reported in Table 6, show that the estimate of the interaction term $\text{Treat}_i \times \text{Post}_{i,t}$ is negative and statistically significant at the 5% level across both models. Its economic magnitude is also large. For example, column (2) of the table shows that the expected default probability decreases by 0.8 percentage points (approximately 6.25% of its standard deviation) after a firm changes to EDGAR filing. The decline in realized stock return volatility is also substantial. Overall, the evidence supports the risk-taking channel of the EDGAR effect.

Table 6. EDGAR Filing and Firm Risk

Independent variable	Stock return volatility (1)	Expected default probability (2)
<i>Treat</i> × <i>Post</i>	−0.012** (−2.19)	−0.008** (−2.21)
<i>Treat</i>	0.002 (0.33)	0.006 (1.41)
<i>logSize</i>	−0.057*** (−17.06)	0.037*** (4.76)
<i>BM</i>	0.069*** (5.96)	−0.015* (−1.75)
<i>Leverage</i>	0.087*** (3.37)	0.163*** (7.69)
<i>Ret</i>	−0.021*** (−6.21)	−0.009 (−1.30)
Industry time trend	Y	Y
Year-quarter FE	Y	Y
Cohort FE	Y	Y
Firm FE	Y	Y
Obs.	19,011	19,762
Adj. R^2	0.693	0.378

Notes. This table examines the effect of EDGAR implementation on firm risk. The analysis uses the propensity score-matched sample. The dependent variable is stock return volatility in column (1) and expected default probability in column (2). All independent variables are defined in the same way as those in Table 3. Robust t -statistics are reported in parentheses and calculated using standard errors clustered at the firm level. FE, fixed effects.

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

5.3. Corporate Governance: Earnings Management

Timely and cheaply available corporate financial information coupled with increased stock price informativeness after EDGAR adoption might strengthen a firm's internal and external control mechanisms. The former is achieved via better alignment between managerial compensation (which is linked to stock prices through stock grants and options) and firm performance, and the latter is achieved via more effective external monitoring. We therefore conjecture that improved corporate governance might be another channel for the documented EDGAR effect.

To investigate this channel, we construct three measures of absolute discretionary accruals— $\text{Accruals}^{\text{Mod Jones}}$, $\text{Accruals}^{\text{DD}}$, and $\text{Accruals}^{\text{MC}}$ —estimated based on the modified Jones (1991), Dechow and Dichev (2002), and McNichols (2002) models, respectively. In these models, discretionary accruals are estimated as the residuals from a regression of actual accruals on a number of economic determinants. Large absolute discretionary accruals are indicative of managers transferring earnings from one period to another.

We then estimate the following regression specification with various earnings management proxies:

$$\begin{aligned} EM_{i,t} = & \alpha_i + \beta_t + \vartheta_g + \delta_1 \text{Treat}_i \times \text{Post}_{i,t} \\ & + \delta_2 \text{Treat}_i + \gamma X_{i,t-1} + \varphi I_c \times t + \varepsilon_{i,t}, \end{aligned} \quad (6)$$

where i indexes firms and t indexes year-quarters. The terms α_i , β_t , ϑ_g , $I_c \times t$, $Treat_{i,t}$, and $Post_{i,t}$ are the same as those in Equation (3). The term EM refers to $Accruals^{Mod_Jones}$, $Accruals^{DD}$, or $Accruals^{MC}$, and $X_{i,t-1}$ represents a vector of firm-level controls. Panel A of Table 7 shows that the coefficient of the interaction term $Treat_i \times Post_{i,t}$ is large and significantly negative across all models. Take the estimate in column (3), for example. It shows that $Accruals^{MC}$ (measured relative to lagged assets) decreases by 23.4 basis points after EDGAR adoption, approximately 7.46% of its standard deviation.

We also use the F -score (i.e., $Fscore$) calculated based on the fitted value of model 1 of Dechow et al. (2011) as an alternative proxy for a firm's earnings management activities. Dechow et al. use the model to predict the likelihood of a firm overstating its earnings and later being subject to SEC enforcement actions for

intentional misrepresentation of results. A higher $Fscore$ indicates a greater likelihood of earnings misstatement by the firm. They also show that a large proportion of earnings manipulators fall in the top quintile of firms ranked by the F -score, so we further define an indicator variable, $HighF_20p$, for this subset of firms. We then estimate Equation (6) but replace $EM_{i,t}$ with $Fscore_{i,t}$ or $HighF_20p_{i,t}$. The results, reported in panel B of Table 7, show that the coefficient of the interaction term $Treated_i \times Post_{i,t}$ is significantly negative at the 5% level across all specifications. The marginal effect reported in column (5) indicates that the likelihood of a firm being classified as an earnings manipulator (i.e., among the top 20% with the highest F -scores) is 2.4 percentage points lower after EDGAR adoption. The overall results lend strong support to the governance channel of the EDGAR effect.

Table 7. EDGAR Filing and Earnings Management

Independent variable	Panel A: Absolute discretionary accruals			Panel B: Accounting misstatement score	
	$Accruals^{Mod_Jones}$ (1)	$Accruals^{DD}$ (2)	$Accruals^{MC}$ (3)	$Fscore$ (4)	$HighF_20p$ (5)
$Treat \times Post$	−0.343** (−2.33)	−0.231*** (−2.93)	−0.234** (−2.00)	−0.030** (−2.52)	−0.024** (−2.32)
$Treat$	0.254* (1.85)	0.055 (0.99)	0.182 (1.62)	0.020* (1.80)	0.014 (1.47)
$logSize$	−0.501 (−1.53)	0.037 (0.21)	0.168 (0.77)	−0.055** (−2.45)	−0.027 (−1.36)
BM	0.133 (0.44)	0.428** (2.21)	−0.376* (−1.75)	−0.195*** (−9.44)	−0.110*** (−6.56)
$Leverage$	−1.134* (−1.65)	−0.448 (−1.18)	−0.795 (−1.28)	0.096* (1.79)	0.068 (1.51)
Age	0.090 (1.32)	0.136*** (3.76)	0.019 (0.39)	0.004 (0.69)	−0.001 (−0.15)
ROA	−3.872 (−1.57)	−5.949*** (−2.84)	−4.125** (−2.31)	0.435*** (3.67)	0.246** (2.47)
$R\&D$	0.365 (0.09)	3.950 (0.73)	−1.503 (−0.43)	−0.324 (−1.41)	−0.021 (−0.10)
$StdCF$	−4.177 (−0.82)	−4.684** (−2.43)	3.377 (1.01)	0.169 (0.50)	−0.018 (−0.08)
$StdSales$	−6.601** (−2.30)	0.496 (0.30)	−1.823 (−0.94)	0.107 (0.50)	0.080 (0.45)
$Tangibility$	−1.770 (−1.19)	−0.161 (−0.21)	−1.017 (−0.88)	−0.297*** (−3.34)	−0.261*** (−3.88)
Industry time trend	Y	Y	Y	Y	Y
Year-quarter FE	Y	Y	Y	Y	Y
Cohort FE	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y
Obs.	13,215	18,159	14,497	21,365	21,365
Adj. R^2	0.327	0.277	0.349	0.606	0.498

Notes. This table reports the effect of EDGAR implementation on a firm's earnings management quality. The analysis uses the propensity score-matched sample. Panel A measures earnings management by the firm's absolute discretionary accruals ($Accruals^{Mod_Jones}$, $Accruals^{DD}$, and $Accruals^{MC}$) based on the modified Jones (1991), Dechow and Dichev (2002), and McNichols (2002) models, respectively. Panel B uses the F -score (i.e., $Fscore$) developed by Dechow et al. (2011) as a measure of the firm's likelihood of earnings manipulation or misstatement. $HighF_20p$ is a dummy variable that takes value 1 if the firm's $Fscore$ is among the top 20% of the sample. The independent variables $Treat_i$, $Post_{i,t}$, $logSize_{i,t-1}$, $BM_{i,t-1}$, $Leverage_{i,t-1}$, and $Ret_{i,t-1}$ are the same as those in Table 3. Firm age ($Age_{i,t-1}$), return on assets ($ROA_{i,t-1}$), R&D to assets ratio ($R\&D_{i,t-1}$), standard deviation of the past 12 quarters' cash flows from operation scaled by total assets ($StdCF_{i,t-1}$), standard deviation of the past 12 quarters' sales scaled by total assets ($StdSales_{i,t-1}$), and asset tangibility ($Tangibility_{i,t-1}$) are included as additional regressors. Robust t -statistics are reported in parentheses and calculated using standard errors clustered at the firm level. FE, fixed effects.

*, **, and *** denote statistical significance the 10%, 5%, and 1% levels, respectively.

6. EDGAR Adoption and Corporate Investment Efficiency

We have thus far documented a significant EDGAR effect on a firm's cost of capital. An interesting question is whether EDGAR adoption has any real effects—in particular, does it affect a firm's investment efficiency? In this section, we employ the methodology suggested by Masulis et al. (2009) to compare the contribution of one extra dollar of capital expenditure and cash to firm value before and after a firm's transition to EDGAR filing.

Specifically, we estimate the following model for the matched pairs of treated and control firms in our sample:

$$\begin{aligned} r_{i,t} - R_{i,t}^B = & \alpha_i + \beta_t + \vartheta_g + \theta_1 \text{Treat}_i \times \text{Post}_{i,t} \times \Delta\text{CAPX}_{i,t} \\ & + \theta_2 \text{Treat}_i \times \text{Post}_{i,t} \\ & + \theta_3 \text{Treat}_i \times \Delta\text{CAPX}_{i,t} + \theta_4 \text{Post}_{i,t} \\ & \times \Delta\text{CAPX}_{i,t} + \theta_5 \text{Treat}_i \\ & + \theta_6 \Delta\text{CAPX}_{i,t} + \varphi I_c \times t + \text{controls} + \varepsilon_{i,t}, \end{aligned} \quad (7)$$

where the dependent variable is the abnormal return of a firm's stock in quarter t , measured by the difference between its raw stock return ($r_{i,t}$) and the benchmark return ($R_{i,t}^B$). The latter is proxied by either $R_{i,t}^{\text{IND}}$

(the value-weighted average return of stock i 's industry) or $R_{i,t}^{\text{Size,BM}}$ (the value-weighted average return of stock i 's matching Fama–French size and book-to-market (BM) portfolio). The capital expenditure in quarter t scaled by total book assets in quarter $t - 1$ is denoted by $\text{CAPX}_{i,t}$, and $\Delta\text{CAPX}_{i,t}$ is the change in capital expenditure from quarter $t - 1$ to t , scaled by total book assets in quarter $t - 1$. The terms α_i , β_t , ϑ_g , Treat_i , $\text{Post}_{i,t}$, and $I_c \times t$ are the same as those in Equation (3). Following Masulis et al. (2009), we control for a list of firm characteristics ($\Delta W_{i,t}$ and $X_{i,t-1}$) that may be correlated with stock returns. The term $\Delta W_{i,t}$ measures the change in W from quarter $t - 1$ to t , which includes earnings before interest and extraordinary items, interest expenses, total dividends, research and development (R&D) expenditures, and total assets net of cash; $X_{i,t-1}$ includes lagged capital expenditure, leverage, and net external financing. Both $\Delta W_{i,t}$ and $X_{i,t-1}$ are scaled by lagged book assets. We also control for the interaction terms between firm characteristics and $\Delta\text{CAPX}_{i,t}$ to mitigate any confounding effect firm-level factors might have on our estimate of θ_1 (deHaan et al. 2021). The estimation uses data from four quarters prior to the event quarter to four quarters afterward, with the event quarter excluded.

Table 8 reports the results. The dependent variable is the quarterly industry-adjusted stock return ($r_{i,t} -$

Table 8. The Marginal Value of Capital Investment

Independent variable	$r - R^{\text{IND}}$			$r - R^{\text{Size, BM}}$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\text{CAPX} \times \text{Treat} \times \text{Post}$	0.933** (2.26)	0.959** (2.33)	0.905** (2.20)	0.907** (2.21)	0.929** (2.28)	0.841** (2.05)
$\text{Treat} \times \text{Post}$	0.000 (0.02)	0.001 (0.07)	0.001 (0.08)	0.007 (0.89)	0.007 (0.95)	0.007 (0.98)
$\Delta\text{CAPX} \times \text{Treated}$	0.001 (0.12)	−0.001 (−0.09)	−0.446* (−1.68)	0.006 (0.92)	0.005 (0.69)	−0.214 (−0.80)
$\Delta\text{CAPX} \times \text{Post}$	−0.500* (−1.94)	−0.512** (−2.00)	−0.481* (−1.80)	−0.327 (−1.25)	−0.335 (−1.28)	−0.300 (−1.11)
Treated	−0.433 (−1.62)	−0.470* (−1.77)	−0.000 (−0.07)	−0.213 (−0.78)	−0.240 (−0.89)	0.005 (0.69)
ΔCAPX	0.417** (2.16)	0.306 (1.48)	0.237 (0.84)	0.137 (0.67)	−0.095 (−0.43)	−0.045 (−0.17)
Firm characteristics	N	N	Y	N	N	Y
Firm characteristics $\times \Delta\text{CAPX}$	N	Y	Y	N	Y	Y
Industry time trend	Y	Y	Y	Y	Y	Y
Year-quarter FE	Y	Y	Y	Y	Y	Y
Cohort FE	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
Obs.	16,735	16,735	16,735	16,902	16,902	16,902
Adj. R^2	0.050	0.054	0.055	0.061	0.065	0.066

Notes. This table examines the impact of EDGAR adoption on the marginal value of a firm's capital expenditure. The analysis uses the propensity score-matched sample. The dependent variable is the abnormal return of a firm's stock in quarter t , measured by the difference between its raw stock return ($r_{i,t}$) and the benchmark return ($R_{i,t}^B$). The benchmark return is proxied by the value-weighted average return of the stock's corresponding industry ($R_{i,t}^{\text{IND}}$) in columns (1)–(3) and by that of the stock's corresponding Fama–French size and book-to-market portfolio ($R_{i,t}^{\text{Size,BM}}$) in columns (4) and (5). The variable of interest is the triple interaction term $\Delta\text{CAPX}_{i,t} \times \text{Treat}_i \times \text{Post}_{i,t}$, where $\Delta\text{CAPX}_{i,t}$ is the change in capital expenditure from quarter $t - 1$ to t , scaled by total book assets in quarter $t - 1$. Robust t -statistics are reported in parentheses and calculated using standard errors clustered at the firm level. FE, fixed effects.

* and ** denote statistical significance the 10% and 5% levels, respectively.

$R_{i,t}^{\text{IND}}$) in columns (1)–(3) and the size/BM adjusted stock return ($r_{i,t} - R_{i,t}^{\text{Size,BM}}$) in columns (4)–(6). The main independent variable of interest is the interaction term $\text{Treat}_i \times \text{Post}_{i,t} \times \Delta \text{CAPX}_{i,t}$. Its coefficient, θ_1 , captures the effect of EDGAR adoption on the marginal value of capital investment, which is estimated by the increase in abnormal stock returns per unit change in a firm's capital expenditure. We find that the estimate of θ_1 is large and significantly positive at the 5% level across all models. Specifically, a one-standard-deviation increase in $\Delta \text{CAPX}_{i,t}$ raises an average firm's quarterly abnormal returns by approximately 1.5 percentage points during the post-EDGAR period relative to control firms.

Next, we examine whether EDGAR implementation has any impact on the marginal value of cash. Following Masulis et al. (2009), we reestimate Equation (7) but replace $\text{CAPX}_{i,t}$ and $\Delta \text{CAPX}_{i,t}$ with $\text{Cash}_{i,t}$ and $\Delta \text{Cash}_{i,t}$. We measure $\text{Cash}_{i,t}$ by the sum of a firm's cash and marketable securities in quarter t , scaled by its total book assets in quarter $t - 1$. The term $\Delta \text{Cash}_{i,t}$ is the change in cash and marketable securities from quarter $t - 1$ to t , scaled by lagged total book assets. Table 9 reports the results. The estimates of the triple interaction term $\text{Treat}_i \times \text{Post}_{i,t} \times \Delta \text{Cash}_{i,t}$ are highly statistically and economically significant across all models.

Take column (5), for example. The point estimate of 0.265 for θ_1 suggests that a one-standard-deviation increase in $\Delta \text{Cash}_{i,t}$ raises an average EDGAR filing firm's abnormal stock returns by approximately 1.8 percentage points per quarter relative to control firms. The overall results indicate that managers are able to create more firm value from their capital expenditure and cash holdings during the post-EDGAR period.

7. Robustness

7.1. Biases in Analyst Forecasts

Gao and Huang (2020) show that analysts' forecasts are more accurate post-EDGAR. If large pre-EDGAR forecast errors generally signal overly optimistic earnings forecasts, our estimates of the cost of capital will be upward biased. We employ three approaches to address this concern.

First, we compute a modified earnings forecast measure, following Chava and Purnanandam (2010):

$$\text{EPS}_{i,t}^{\text{modified}} = x_i \text{LowEPS}_{i,t} + (1 - x_i) \text{HighEPS}_{i,t}, \quad (8)$$

where $\text{LowEPS}_{i,t}$ and $\text{HighEPS}_{i,t}$ denote the lowest and the highest earnings forecasts for the firm-quarter, respectively, and x_i denotes the percentile ranking of firm i 's consensus one-year-ahead signed earnings forecast error

Table 9. The Marginal Value of Cash

Independent variable	$r - R^{\text{IND}}$			$r - R^{\text{Size, BM}}$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \text{Cash} \times \text{Treat} \times \text{Post}$	0.294** (2.09)	0.293** (2.12)	0.341** (2.52)	0.268** (2.09)	0.265** (2.08)	0.305** (2.37)
$\text{Treat} \times \text{Post}$	0.008 (1.16)	0.008 (1.12)	0.008 (1.11)	0.013* (1.96)	0.013* (1.93)	0.013* (1.93)
$\Delta \text{Cash} \times \text{Treat}$	−0.006 (−1.08)	−0.007 (−1.15)	−0.121 (−1.26)	0.002 (0.39)	0.002 (0.32)	−0.081 (−0.88)
$\Delta \text{Cash} \times \text{Post}$	−0.155 (−1.47)	−0.167* (−1.66)	−0.194** (−1.99)	−0.091 (−0.97)	−0.098 (−1.07)	−0.123 (−1.30)
Treat	−0.120 (−1.11)	−0.113 (−1.11)	−0.007 (−1.17)	−0.084 (−0.89)	−0.077 (−0.84)	0.002 (0.29)
ΔCash	0.156* (1.75)	0.193** (2.24)	0.182* (1.86)	0.099 (1.32)	0.128* (1.69)	0.119 (1.24)
Firm characteristics	N	N	Y	N	N	Y
Firm characteristics $\times \Delta \text{Cash}$	N	Y	Y	N	Y	Y
Industry time trend	Y	Y	Y	Y	Y	Y
Year-quarter FE	Y	Y	Y	Y	Y	Y
Cohort FE	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y
Obs.	17,918	17,918	17,918	18,088	18,088	18,088
Adj. R^2	0.041	0.047	0.048	0.056	0.061	0.062

Notes. This table examines the impact of EDGAR adoption on the marginal value of a firm's cash. The analysis uses the propensity score-matched sample. The dependent variable is the abnormal return of a firm's stock in quarter t , measured by the difference between its raw stock return ($r_{i,t}$) and the benchmark return ($R_{i,t}^{\text{B}}$). The benchmark return is proxied by the value-weighted average return of the stock's corresponding industry ($R_{i,t}^{\text{IND}}$) in columns (1)–(3) and by that of the stock's corresponding Fama–French size and book-to-market portfolio ($R_{i,t}^{\text{Size,BM}}$) in columns (4)–(6). The variable of interest is the triple interaction term $\Delta \text{Cash}_{i,t} \times \text{Treat}_i \times \text{Post}_{i,t}$, where $\Delta \text{Cash}_{i,t}$ is the change in cash and marketable securities from quarter $t - 1$ to t , scaled by total book assets in quarter $t - 1$. Robust t -statistics are reported in parentheses and calculated using standard errors clustered at the firm level. FE, fixed effects.

* and ** denote statistical significance at the 10% and 5% levels, respectively.

(defined as the consensus forecasted earnings less actual earnings, divided by lagged total assets) among all firms in quarter $t - 1$, with x_i ranging between 0 and 1. We then use this modified EPS to estimate the ICC models. We label the resultant average cost of equity as $R_{i,t}^{FE}$.

Second, prior studies (e.g., Rajan and Servaes 1997) show that analysts exhibit optimism bias toward firms with large investment banking businesses. To adjust for this potential bias, we compute a modified earnings forecast measure according to Equation (8) but with x_i defined as the percentile ranking of a firm's net external financing during the year preceding quarter t . We calculate a firm's net external financing based on the method suggested by Chava and Purnanandam (2010). The average cost of equity capital estimated using this modified earnings forecast measure is labeled $R_{i,t}^{ExtFin}$.

Third, the literature (e.g., Michaley and Womack 1999) documents that analysts affiliated with investment banks are likely to exhibit bias in favor of the banks' clients. We classify an analyst as affiliated if the brokerage house he or she works for is an underwriter of the covered firm's initial public offerings (IPOs) in the past five years, seasoned equity offerings (SEOs) in the past two years, or bond issues in the past year. We then use the median forecasts issued by unaffiliated analysts in the estimation of the ICC models and label the estimated cost of equity as $R_{i,t}^{Unaffiliated}$.

We then estimate Equation (3) with $R_{i,t}^{FE}$, $R_{i,t}^{ExtFin}$, or $R_{i,t}^{Unaffiliated}$ as the dependent variable. The results, reported in Table 10, show that the EDGAR effect remains strong in all three cases.

7.2. Pretrend and Falsification Tests

Our identification strategy assumes that, absent EDGAR implementation, there will be no significant difference in the cost of equity capital between treated and control firms. To examine this identifying assumption, we define two additional dummy variables: $Post(-j)$, $j = 1$ or 2 , takes value 1 in the j th quarter before a firm switches to EDGAR filing and 0 otherwise. In Table 11, column (1), we add two additional interaction terms, $Treat_i \times Post_{i,t}(-2)$ and $Treat_i \times Post_{i,t}(-1)$. The results show that the estimates of the two pre-event interaction terms are both insignificant, suggesting no discernible difference in the cost of capital between treated and control firms prior to EDGAR adoption.

We also implement three falsification tests. In the first, we define pseudo-event dates (April 1991–May 1994) of EDGAR filers as two years prior to the actual event dates (April 1993–May 1996). In the second, we set pseudo-event dates (April 1989–May 1992) as four years prior to the actual dates (April 1993–May 1996) so that the former will not overlap with the latter. In the third, we set pseudo-event dates (April 1995–May 1998) as two years after the actual dates. We then reestimate Equation (3) using these pseudo-event dates.¹³

Table 10. Bias in Analyst Forecasts

Independent variable	$R_{i,t}^{FE}$ (1)	$R_{i,t}^{ExtFin}$ (2)	$R_{i,t}^{Unaffiliated}$ (3)
$Treat \times Post$	−0.309** (−2.22)	−0.299** (−2.15)	−0.320** (−2.36)
$Treat$	−0.007 (−0.06)	−0.039 (−0.31)	0.083 (0.64)
$logSize$	−0.642* (−1.86)	−1.049*** (−2.95)	−0.733** (−2.18)
BM	1.697*** (4.32)	1.601*** (4.01)	1.575*** (4.14)
$Leverage$	4.659*** (3.51)	4.777*** (3.62)	4.673*** (3.62)
Ret	0.083* (1.65)	0.155*** (3.05)	0.090** (1.97)
$StdRet$	1.867*** (2.98)	2.245*** (3.45)	1.979*** (3.17)
Industry time trend	Y	Y	Y
Year-quarter FE	Y	Y	Y
Cohort FE	Y	Y	Y
Firm FE	Y	Y	Y
Obs.	15,500	15,500	15,500
Adj. R^2	0.789	0.781	0.822

Notes. This table reestimates Equation (3) but with ICC models estimated based on three alternative modified EPS measures. The first two measures are calculated using the following formula (Equation (8)):

$$EPS_{i,t}^{modified} = x_i LowEPS_{i,t} + (1 - x_i) HighEPS_{i,t},$$

where $LowEPS_{i,t}$ and $HighEPS_{i,t}$ denote the lowest and the highest earnings forecast available for firm i in quarter t , respectively. In column (1), we define x_i as the percentile ranking of the firm's consensus one-year-ahead signed earnings forecast errors in quarter $t - 1$. The average cost of equity capital estimated using this modified earnings forecast measure is labeled as $R_{i,t}^{FE}$. In column (2), we define x_i as the percentile ranking of the firm's net external financing over the past year and label the estimated average cost of equity capital as $R_{i,t}^{ExtFin}$. In column (3), we remove forecasts made by affiliated analysts and use the median forecast made by unaffiliated analysts as the consensus forecast. The resulting estimate of the average cost of equity is labeled $R_{i,t}^{Unaffiliated}$. The independent variables are defined in the same way as those in Table 3. Robust t -statistics are reported in parentheses and calculated using standard errors clustered at the firm level.

*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Columns (2)–(4) of Table 11 report the results. In all three tests, the estimates of $Treated_i \times Post_{i,t}$ are insignificantly different from 0.

7.3. Dynamic Treatment Effects

So far, we have not considered the possibility of heterogeneous treatment effects over time. Baker et al. (2022), Cengiz et al. (2019), and Callaway and Sant'Anna (2021) suggest that in the presence of dynamic treatment effects, bias can arise if early-treatment groups are used as control firms for late-treatment groups. We address this concern in two ways.

First, we follow Cengiz et al. (2019) to conduct a robustness test using only those firms that have never been treated before as our control firms. They show

Table 11. Robustness: Pretrend, Falsification Tests, Two-Way Clustering, Interim Dummy, and Alternative Cost of Capital Measures

Independent variable	Pretrend (1)	Falsification test I (2)	Falsification test II (3)	Falsification test III (4)	R^{PSS} (5)	R^{PEG} (6)	Cluster at firm and time (7)	Include Interim dummy (8)
$Treat \times Post$	−0.454*** (3.20)	−0.212 (−0.97)	0.122 (0.80)	−0.164 (−0.54)	−0.681*** (−2.68)	−0.537*** (−3.53)	−0.395** (2.95)	−0.425*** (3.14)
$Treat \times Post$ (−2)	−0.110 (0.70)							
$Treat \times Post$ (−1)	−0.119 (1.28)							
$Treat \times$ <i>Interim</i>								−0.153 (0.90)
Firm-level controls	Y	Y	Y	Y	Y	Y	Y	Y
Industry time trend	Y	Y	Y	Y	Y	Y	Y	Y
Year-quarter FE	Y	Y	Y	Y	Y	Y	Y	Y
Cohort FE	Y	Y	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y
Obs.	15,500	9,635	10,850	3,051	13,152	8,939	15,500	15,500
Adj. R^2	0.845	0.810	0.841	0.857	0.792	0.830	0.845	0.845

Notes. This table reports the results of several robustness tests using the propensity score-matched sample. In column (1), we reestimate Equation (3) with two additional regressors, $Treat_i \times Post_{i,t}(-2)$ and $Treat_i \times Post_{i,t}(-1)$, where $Post(-j)$, $j = 1$ or 2 , takes value 1 in the j th quarter before a firm switches to EDGAR filing and 0 otherwise. In columns (2)–(4), we report the results of three falsification tests. In the first, we define pseudo-event dates (April 1991–May 1994) for the eight cohorts of EDGAR filers as two years prior to the actual event dates (April 1993–May 1996). We then reestimate Equation (3) with the $Post_{i,t}$ dummy redefined based on this pseudo-event schedule. In the second test, reported in column (3), we set pseudo-event dates (April 1989–May 1992) as four years prior to the actual dates (April 1993–May 1996). In the third test, reported in column (4), we set pseudo-event dates (April 1995–May 1998) as two years after the actual dates. In columns (5) and (6), we reestimate Equation (3) with two alternative measures of the implied cost of capital, R^{PSS} and R^{PEG} . The former is estimated based on the approach proposed by Pastor et al. (2008), and the latter is based on the PEG ratio model. In column (7), we reestimate Equation (3) with standard errors clustered at the firm and year-quarter levels. In column (8), we redefine the event month as January 1994 for firms that filed in EDGAR before 1994. For these firms, the modified $Post_{i,t}$ variable takes value 1 from Q2 1994 onward and 0 otherwise. In addition, we create an *Interim* indicator for these firms during this interim period. Except for columns (5) and (6), the dependent variable is $COC_{i,t}$. Robust t -statistics are reported in parentheses and calculated using standard errors clustered at the firm level. FE, fixed effects.

** and *** denote statistical significance at the 5% and 1% levels, respectively.

that the use of “clean” control firms helps address the issue of heterogeneous treatment effects in a staggered treatment design. In our setting, the pool of clean control firms consists of those firms that have not been treated until after quarter $t + 4$, with t being the event quarter. We include only those firms in EDGAR phase-in groups 2–5 as our treated firms. For each treated firm, we identify a matching clean control firm using the estimated propensity scores from a logit model to perform one-to-one nearest-neighbor matching with replacement. We then reestimate Equation (3) using this smaller sample of treated firms and their matched clean control firms, with an eight-quarter event window.¹⁴ The result is reported in column (1) of Table 12. The estimate of the interaction term $Treat_i \times Post_{i,t}$ is -0.477 (t -stat = -3.01), which is qualitatively similar to the estimate of -0.395 (t -stat = -3.19) from our baseline model, reported in column (3) of panel B of Table 3.

Second, Callaway and Sant’Anna (2021) propose a DiD test methodology that is unbiased and consistent in the presence of heterogeneous treatment effects in multiple treatment events and periods. We follow their approach to estimate the cost of capital effect of EDGAR implementation. Specifically, we again include only those firms in EDGAR phase-in groups 2–5 as our treated firms. For each cohort of treated firms at quarter $t + k$ (with t being the event quarter and k ranging from 1 to 4), we estimate their average treatment effect using those firms still not yet treated at the time as the comparison (or control) group. To ensure that the covariates of treated and control firms are balanced, control firms that have attributes more (less) similar to firms in the treated group receive more (less) weight, with similarity measured based on a propensity score model. We then calculate the aggregate treatment effect using the simple aggregation procedure from equation (3.10) in their paper, which gives more weight to groups with larger group

Table 12. Robustness: Alternative Estimation Methods

Independent variable	Cengiz et al. (2019) (1)	Callaway and Sant’Anna (2021)	
		Unconditional (2)	Conditional (3)
$Treat_i \times Post_{i,t}$	−0.477*** (−3.01)	−0.431*** (−3.60)	−0.482*** (−4.03)

Notes. This table reports the average cost of capital effect of EDGAR adoption using two alternative estimation methods based on Cengiz et al. (2019) and Callaway and Sant’Anna (2021). In these tests, we include only those firms in EDGAR phase-in groups 2–5 as our treated firms. In column (1), we report the result based on Cengiz et al. (2019). Specifically, for each treated firm, we identify a matching clean control firm using the estimated propensity scores from a logit model. We then reestimate Equation (3) with this smaller sample of matched pairs of firms. The estimate of the interaction term $Treat_i \times Post_{i,t}$ is reported in the main entry, and the robust t -statistic calculated using standard errors clustered at the firm level is reported in parentheses. The number of firm-quarter observations and adjusted R^2 are 11,249 and 0.857, respectively. In columns (2) and (3), we report the estimation results based on Callaway and Sant’Anna (2021). Column (2) assumes unconditional parallel trends, and column (3) assumes that parallel trends hold between treated and control firms after conditioning on covariates (i.e., the variables listed in Table 3, panel A in our setting). The estimates of the cost of capital effect of EDGAR adoption are reported in the main entries, and the t -statistics calculated based on the bootstrap procedure described in Callaway and Sant’Anna (2021) are reported in parentheses. The number of observations is 19,432 firm-quarters.

*** denotes statistical significance at the 1% level.

sizes. Following Callaway and Sant’Anna (2021), we report two versions of the estimated average treatment effect: unconditional and conditional. The former (reported in column (2) of Table 12) assumes unconditional parallel trends, and the latter (reported in column (3)) assumes that parallel trends hold between treated and control firms after conditioning on covariates (i.e., the variables listed in Table 3, panel A in our setting). The estimated cost of capital effects of EDGAR adoption are -0.431 (t -stat = -3.6) and -0.482 (t -stat = -4.03) under the assumption of unconditional and conditional parallel trends, respectively. These estimates are again qualitatively similar to the estimate of -0.395 from our baseline model reported in Table 3.

7.4. Other Robustness Tests

We conduct several additional robustness tests in this subsection. First, we estimate two alternative measures of the implied cost of capital, R^{PSS} and R^{PEG} . The former (R^{PSS}) is estimated based on Pástor et al. (2008), which is similar to that proposed by Gebhardt et al. (2001) but with a few different assumptions. The latter (R^{PEG}) is estimated based on the price-earnings growth (PEG) ratio model. We describe the two models in detail in Appendix B. We reestimate Equation (3) with these two alternative measures of the cost of capital and report the results in Table 11, columns (5) and (6). We still find a highly economically and statistically significant EDGAR effect.

Second, we reestimate Equation (3) with standard deviations clustered at the firm and year-quarter levels. We find that the interaction term $Treat_i \times Post_{i,t}$ remains highly significant at the 1% level, as reported in column (7) of Table 11.

Third, electronic EDGAR filings were initially accessible only through a commercial data vendor and became freely available on January 17, 1994. Given that

the initial data subscription fee might have deterred some investors from accessing EDGAR filings, we redefine the event month as January 1994 for firms that filed in EDGAR before 1994. For these firms, the modified $Post_{i,t}$ variable takes value 1 from Q2 1994 onward and 0 otherwise. In addition, we create an *Interim* indicator for these firms, which takes value 1 over the period from Q3 1993 to Q1 1994 and 0 otherwise. We then reestimate Equation (3) with the modified $Post_{i,t}$ variable plus two additional terms, $Interim_{i,t}$ and $Treat_i \times Interim_{i,t}$. Column (8) of Table 11 shows that the estimate of $Treat_i \times Interim_{i,t}$ is negative but statistically insignificant, whereas the estimate of $Treat_i \times Post_{i,t}$ is economically large and highly significant at the 1% level.

Last, we examine whether the cost of capital effect occurs only after a firm begins filing in EDGAR. To this end, we create two dummy variables: $F_Post_{i,m}$ and $NF_Post_{i,m}$. The former (respectively, latter) takes value 1 if firm i is subject to mandatory EDGAR filing and has (respectively, has not) filed its first filing in EDGAR by year-month m .¹⁵ We then reestimate Equation (4) with $Post_{i,m}$ replaced by these two new dummy variables. For all three measures of stock illiquidity, the estimate of the interaction term $Treat_i \times F_Post_{i,m}$ is highly significantly negative at the 1% level. Although the coefficient of the term $Treat_i \times NF_Post_{i,m}$ is also negative, it is relatively small in magnitude and statistically insignificant. Our results thus suggest that the bulk of the EDGAR effect occurs only after a firm actually files in EDGAR.

8. Conclusion

The effects of RegTech adoption on a firm’s capital allocation and investment efficiency have so far not received significant attention in the literature. This study focuses on the first major RegTech event in recent times, the implementation of EDGAR in the period 1993–1996. The

EDGAR system was introduced to make corporate filings more easily accessible to financial market participants and to generate more informed securities markets (SEC 1993). We find evidence that, aside from the expected securities markets effect, EDGAR adoption has a substantial impact on a firm's risk, stock liquidity, governance, and cost of capital. In addition, the marginal values of a firm's capital investment and cash also increase significantly post-EDGAR.

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Appendix B. Implied Cost of Equity Capital Models

ICC model	Formula and assumptions
Gebhardt et al. (2001)	$P_{\tau} = BV_{\tau} + \sum_{k=1}^{11} \frac{(\hat{x}_{\tau+k} - R \cdot BV_{\tau+k-1})}{(1+R)^k} + \frac{(\hat{x}_{\tau+12} - R \cdot BV_{\tau+11})}{R(1+R)^{11}},$ where P_{τ} is the stock price of a firm at date τ, R is the implied cost of equity capital, BV_{τ} is the book equity per share at date τ, and $BV_{\tau+k-1}$ is the expected book equity per share at [(date τ) + ($k-1$) years]. The term $\hat{x}_{\tau+k}$ is the expected future earnings per share for the period [(date τ) + ($k-1$) years, (date τ) + k years]. The terms $\hat{x}_{\tau+1}$ and $\hat{x}_{\tau+2}$ are obtained directly from median analyst one-year- and two-year-ahead earnings per share forecasts in I/B/E/S, respectively, and $\hat{x}_{\tau+3}$ is estimated by the relation: $\hat{x}_{\tau+3} = \hat{x}_{\tau+2} \cdot (1 + l_{tg})$, where l_{tg} is the forecast of long-term earnings growth from I/B/E/S. After the first three years, the forecasted return on equity (i.e., $\hat{x}_{\tau+k}/BV_{\tau+k-1}$) is assumed to revert linearly to the historical sector-specific median return on equity (ROE) by year 11 and remains constant afterward. We identify each sector by its two-digit Standard Industrial Classification code and compute the median ROE using data over the past 10 years for all firms with nonnegative earnings. The expected future book equity is calculated based on clean surplus accounting: $B_{\tau+k} = B_{\tau+k-1} + \hat{x}_{\tau+k} - \hat{d}_{\tau+k}$, where $\hat{d}_{\tau+k}$ is the expected dividend per share at time $\tau+k$. The dividend payout ratio is estimated as the current dividends divided by the current earnings for firms with positive earnings or as current dividends divided by $(0.06 \times \text{total assets})$ for firms with negative earnings. We use a numerical approximation program to solve for the implied cost of capital R. This model represents a special case of the residual income valuation model.
Claus and Thomas (2001)	$P_{\tau} = BV_{\tau} + \sum_{k=1}^5 \frac{(\hat{x}_{\tau+k} - R \cdot BV_{\tau+k-1})}{(1+R)^k} + \frac{(\hat{x}_{\tau+5} - R \cdot BV_{\tau+4})(1+g)}{(R-g)(1+R)^5},$ where P_{τ}, BV_{τ}, $BV_{\tau+k-1}$, $\hat{x}_{\tau+1}$, and $\hat{x}_{\tau+2}$ are the same as those defined in the description of the Gebhardt et al. (2001) model. The term $\hat{x}_{\tau+k}$, $k=3$ to 5, is estimated by the relation $\hat{x}_{\tau+k} = \hat{x}_{\tau+k-1} \cdot (1 + l_{tg})$, where l_{tg} is the forecast of long-term earnings growth from I/B/E/S. After the first five years, the nominal residual income $(\hat{x}_{\tau+k} - R \cdot BV_{\tau+k-1})$ is assumed to grow at the rate g, which is set to the contemporaneous (annualized) risk-free rate minus 3%. Then, we use a numerical approximation program to solve for the implied cost of capital R. This model represents a special case of the residual income valuation model.
Ohlson and Juettner-Nauroth (2005)	$P_{\tau} = \frac{\hat{x}_{\tau+1}}{R} \cdot \frac{(g_{ST} - g_{LT} + R \cdot \hat{d}_{\tau+1}/\hat{x}_{\tau+1})}{(R - g_{LT})},$ where P_{τ}, $\hat{x}_{\tau+1}$, and $\hat{d}_{\tau+1}$ are the same as those defined in the description of the Gebhardt et al. (2001) model. Following Gode and Mohanram (2003), we set the short-term growth rate g_{ST} as the average between the percentage change from one-year-ahead to two-year-ahead earnings forecast and the five-year-ahead earnings growth forecast provided by financial analysts in the I/B/E/S database. The growth in abnormal earnings per share beyond the first year is referred to as the long-term growth rate g_{LT}, which is assumed to equal the contemporaneous (annualized) risk-free rate minus 3%. The implementation of this model requires a positive change in forecasted earnings to yield a numerical solution. We use a numerical approximation program to solve for the implied cost of capital R. This model represents a special case of the abnormal earnings growth valuation model proposed by Ohlson and Juettner-Nauroth (2005).
Modified PEG ratio model by Easton (2004)	$P_{\tau} = \frac{(\hat{x}_{\tau+2} - \hat{x}_{\tau+1} + R \cdot \hat{d}_{\tau+1})}{R^2},$ where P_{τ}, $\hat{x}_{\tau+1}$, $\hat{x}_{\tau+2}$, and $\hat{d}_{\tau+1}$ are the same as those defined in the description of the Gebhardt et al. (2001) model. The implementation of this model requires a positive change in forecasted earnings to yield a numerical solution. We use a numerical approximation program to solve for the implied cost of capital R. This model is also a special case of the abnormal earnings growth valuation model proposed by Ohlson and Juettner-Nauroth (2005).

Appendix A. EDGAR Phase-in Schedule

This appendix documents the phase-in schedule for the SEC's EDGAR system for each group of firms according to SEC Release No. 33-6977 (February 23, 1993) [58 FR 14628].

- April 26, 1993: Phase-in of Group 1.
- July 19, 1993: Phase-in of Group 2.
- October 4, 1993: Phase-in of Group 3.
- December 6, 1993: Phase-in of Group 4.
- August 1994: Phase-in of Group 5.
- November 1994: Phase-in of Group 6.
- May 1995: Phase-in of Group 7.
- August 1995: Phase-in of Group 8.
- November 1995: Phase-in of Group 9.
- May 1996: Phase-in of Group 10.

Appendix B. (Continued)

ICC model	Formula and assumptions
PEG ratio by Botosan and Plumlee (2005)	$P_{\tau} = \frac{(\hat{x}_{\tau+5} - \hat{x}_{\tau+4})}{R^2},$ where P_{τ} is the stock price of a firm at date τ, R is the implied cost of equity capital, and $\hat{x}_{\tau+k}$ is the expected future earnings per share for the period $[(\text{date } \tau) + (k - 1) \text{ years}, (\text{date } \tau) + k \text{ years}]$. The terms $\hat{x}_{\tau+1}$ and $\hat{x}_{\tau+2}$ are obtained directly from median analyst one-year- and two-year ahead earnings per share forecasts in I/B/E/S. The term $\hat{x}_{\tau+k}$, $k = 3$ to 5, is estimated by the relation $\hat{x}_{\tau+k} = \hat{x}_{\tau+k-1} \cdot (1 + ltg)$, where ltg is the forecast of long-term earnings growth from I/B/E/S.
Pástor et al. (2008)	$P_{\tau} = \sum_{k=1}^{15} \frac{\hat{x}_{\tau+k}(1 - b_{\tau+k})}{(1+R)^k} + \frac{\hat{x}_{\tau+16}}{R(1+R)^{15}},$ where P_{τ} is the stock price of a firm at date τ, and R is the implied cost of equity capital. The term $\hat{x}_{\tau+k}$ is the expected future earnings per share for the period $[(\text{date } \tau) + (k - 1) \text{ years}, (\text{date } \tau) + k \text{ years}]$. The terms $\hat{x}_{\tau+1}$ and $\hat{x}_{\tau+2}$ are obtained directly from median analyst one-year- and two-year-ahead earnings per share forecasts in I/B/E/S, respectively, and $\hat{x}_{\tau+3}$ is estimated by the relation $\hat{x}_{\tau+3} = \hat{x}_{\tau+2} \cdot (1 + ltg)$, where ltg is the forecast of long-term earnings growth from I/B/E/S. Firms with growth rates above 100% (below 2%) are assigned values of 100% (2%). The plowback rate for the first two years ($b_{\tau+k}$, where $k = 1, 2$) is assumed to be 1 minus the firm's most recent net payout ratio (which is the sum of dividends and share repurchases minus any issuance of new stock, scaled by net income). The forecasted earnings $\hat{x}_{\tau+k}$ for $k = 4$ to 16 are calculated as $\hat{x}_{\tau+k} = \hat{x}_{\tau+k-1} \cdot (1 + g_{\tau+k})$, with $g_{\tau+k} = g_{\tau+k-1} \cdot \exp[\log(g/g_{\tau+3})/(T - 1)]$. We set $g_{\tau+3}$ as ltg or as $(\hat{x}_{\tau+2}/\hat{x}_{\tau+1}) - 1$ if ltg is not available. The plowback rate $b_{\tau+k}$ for $k = 3$ to 15 is calculated as $b_{\tau+k} = b_{\tau+k-1} - \frac{b_{\tau+2} - b}{14}$. That is, after the first three years, the earnings growth rate is assumed to revert linearly to the long-run nominal gross domestic product (GDP) growth rate g by year $\tau + 17$ and remains constant afterward. In addition, after the first two years, the plowback rate is assumed to revert linearly to a steady-state value b by year $\tau + 16$. Denote by g the sum of the long-run real GDP growth rate and the long-run average rate of inflation, and $b = g/R$.

Notes. This table describes the six implied cost of equity capital models we use in this paper. We estimate these models at the end of each quarter τ from Q2 1992 to Q2 1997.

Appendix C. Variable Definitions

Variable	Definition
Quarterly frequency	
COC (%)	Cost of equity in excess of the risk-free rate (in percent). The cost of equity is proxied by the equal-weighted average of the ICC estimates from four different models: Gebhardt et al. (2001), Claus and Thomas (2001), Ohlson and Juettner-Nauroth (2005), and Easton (2004). The risk-free rate is the annualized one-month T-bill rate. (Sources: CRSP and I/B/E/S)
R^{PC} (%)	First principal component of the four ICC estimates. (Sources: CRSP and I/B/E/S)
$R^{Model \text{ fit}}$ (%)	Weighted average of the four ICC estimates in excess of the risk-free rate (in percent), where the weight is the regression R^2 of each individual ICC model. (Sources: CRSP and I/B/E/S)
R^{FE} (%)	Forecast error-adjusted cost of equity in excess of the risk-free rate (in percent). We first compute a modified earnings forecast measure using Equation (8), in which x_i is the percentile ranking of firm i 's consensus one-year-ahead signed earnings forecast error in the previous quarter. We then use this modified earnings forecast to estimate the four ICC models. R^{FE} is the weighted average of the four ICC estimates in excess of the risk-free rate. (Sources: CRSP and I/B/E/S)
R^{ExtFin} (%)	External financing-adjusted cost of equity in excess of the risk-free rate (in percent). It is calculated in the same way as R^{FE} except that x_i is the percentile ranking of firm i 's net external financing over the past year. (Sources: CRSP, I/B/E/S, and Compustat)
$R^{Unaffiliated}$ (%)	Cost of equity in excess of the risk-free rate (in percent), calculated based on forecasts made by unaffiliated analysts only. It is computed in the same way as COC, except that the consensus analyst forecast is replaced by the median forecast made by unaffiliated analysts. We classify an analyst as affiliated if the brokerage house she works for is an underwriter for the covered firm's IPO in the past five years, SEOs in the past two years, or bond issues in the past year. (Sources: CRSP and I/B/E/S)
PostEDGAR	A dummy variable that takes value 1 if firm i is subject to mandatory EDGAR filing by year-quarter t and 0 otherwise.
Treat	A dummy variable for treated firms.
Post	A dummy variable for the postevent period.
logSize	Natural logarithm of total book assets. (Source: Compustat)
BM	Ratio of book value to market value of equity. (Source: Compustat)
Leverage	Total debt divided by total book assets. (Source: Compustat)
Ret	Stock returns during the quarter. (Source: CRSP)

Appendix C. (Continued)

Variable	Definition
<i>FBias</i>	One-year-ahead analyst forecast error. It is the difference between analysts' consensus one-year-ahead earnings forecast and actual earnings, divided by total assets. (Sources: CRSP, Compustat, and I/B/E/S)
<i>Age</i>	Number of years since a firm first appeared in the Compustat database. (Source: Compustat)
<i>Sales growth</i>	Growth of quarterly sales from four quarters earlier to the current quarter. (Source: Compustat)
<i>Tangibility</i>	Net property, plant, and equipment divided by total assets. (Source: Compustat)
<i>IO</i>	Number of shares held by institutional investors relative to the total number of shares outstanding. (Sources: Thomson-Reuters 13F and CRSP)
<i>R&D</i>	R&D expenditure scaled by total assets. (Source: Compustat)
<i>ROA</i>	Ratio of net income to total assets. (Source: Compustat)
<i>StdCF</i>	Standard deviation of the past 12 quarters' cash flows from operations scaled by total assets. (Source: Compustat)
<i>StdSales</i>	Standard deviation of the past 12 quarters' sales scaled by total asset. (Source: Compustat)
<i>Stock return volatility (StdRet)</i>	Annualized stock return volatility. It is estimated by the standard deviation of the daily stock returns during the quarter multiplied by the square root of 252. If there are fewer than 20 nonmissing daily returns in a given quarter, we set it as a missing value. (Source: CRSP)
<i>Expected default probability</i>	The probability that the value of a firm will fall below the face value of debt at the one-year forecasting horizon. Specifically, we follow the procedure described in Bharath and Shumway (2008) to estimate the following two nonlinear equations under the Merton distance to default model: $(1) E = VN(d1) - d^{-rT}FN(d2), \text{ where } d1 = \frac{\ln(\frac{V}{F}) + (r + 0.5\sigma_V^2)T}{\sigma_V\sqrt{T}}, d2 = d1 - \sigma_V\sqrt{T}.$ Here, E and V refer to the market value of the firm's equity and assets, respectively; F is the face value of debt; σ_V is the volatility of assets; and r is the risk-free rate. Denote by $N(\cdot)$ the cumulative standard normal distribution function: $(2) \sigma_E = \left(\frac{V}{E}\right)N(d1)\sigma_V - d^{-rT}FN(d2), \text{ where } \sigma_E \text{ denotes the volatility of the firm's equity.}$ We set the forecasting horizon T to one year and use the observed values of a firm's E, σ_E, F, and r as the input to solve for σ_V.
<i>Accruals^{Mod_Jones}</i>	Absolute value of discretionary accruals scaled by the previous quarter's total assets, where the discretionary accruals are estimated based on the modified Jones (1991) model. We follow a similar approach used by Shivakumar (2000) and Armstrong et al. (2013) to estimate expected accruals. Specifically, for each two-digit Standard Industrial Classification industry-quarter with at least 15 observations, we estimate the cross-sectional version of the modified Jones model as follows: $\frac{ac_{i,t}}{A_{i,t-1}} = \beta_1 + \beta_2[(\Delta rev_{i,t} - \Delta rec_{i,t})/A_{i,t-1}] + \beta_3(ppe_{i,t}/A_{i,t-1}) + \varepsilon_{i,t}, \quad (C.1)$ where $ac_{i,t}$ denotes total accruals of firm i in quarter t. We calculate total accruals as the difference between earnings before extraordinary items and cash flows from operations. $A_{i,t-1}$ is the total assets for the firm at the end of quarter $t - 1$; $\Delta rev_{i,t}$ and $\Delta rec_{i,t}$ refer to the change in net sales and the change in accounts receivable, respectively, from quarter $t - 1$ to quarter t; $ppe_{i,t}$ represents the gross property, plant, and equipment at the end of quarter t; and $\varepsilon_{i,t}$ is the error term. We then estimate expected accruals for firm i in quarter t as $\hat{E}_{i,t}^{Mod_Jones} = \hat{\beta}_1 + \hat{\beta}_2[(\Delta rev_{i,t} - \Delta rec_{i,t})/A_{i,t-1}] + \hat{\beta}_3(ppe_{i,t}/A_{i,t-1}), \quad (C.2)$ where $\hat{\beta}_1$, $\hat{\beta}_2$, and $\hat{\beta}_3$ are the OLS estimates of β_1, β_2, and β_3, respectively. The absolute discretionary accruals are calculated as $Accruals_{i,t}^{Mod_Jones} = \left \frac{ac_{i,t}}{A_{i,t-1}} - \hat{E}_{i,t}^{Mod_Jones} \right . \quad (C.3)$
<i>Accruals^{DD}</i>	Absolute value of discretionary accruals scaled by the previous quarter's total assets, where the discretionary accruals are estimated based on the Dechow and Dichev (2002) model. It is calculated using a similar approach as that for <i>Accruals^{Mod_Jones}</i> except that we replace Equation (C.1) with Equation (C.4). The absolute discretionary accruals (<i>Accruals^{DD}</i>) are the absolute difference between the actual accruals and the estimated expected accruals, with the latter calculated as the fitted value of Equation (C.4): $\frac{ac_{i,t}}{A_{i,t-1}} = \theta_1 + \theta_2[(CFO_{i,t-1})/A_{i,t-1}] + \theta_3(CFO_{i,t}/A_{i,t-1}) + \theta_4(CFO_{i,t+1}/A_{i,t-1}) + \varepsilon_{i,t}, \quad (C.4)$ where $CFO_{i,t}$ denotes cash flow from operations for firm i in quarter t.
<i>Accruals^{MC}</i>	Absolute value of discretionary accruals scaled by the previous quarter's total assets, where the discretionary accruals are estimated based on the McNichols (2002) model. It is calculated using a similar approach as that for <i>Accruals^{Mod_Jones}</i> except that we replace Equation (C.1) with Equation (C.5). The absolute

Appendix C. (Continued)

Variable	Definition
	discretionary accruals ($Accruals^{MC}$) are the absolute difference between the actual accruals and the estimated expected accruals, with the latter calculated as the fitted value of Equation (C.5):
	$\frac{ac_{i,t}}{A_{i,t-1}} = \gamma_1 + \gamma_2[(CFO_{i,t-1})/A_{i,t-1}] + \gamma_3\left(\frac{CFO_{i,t}}{A_{i,t-1}}\right) + \gamma_4\left(\frac{CFO_{i,t+1}}{A_{i,t-1}}\right) + \gamma_5[\Delta rev_{i,t}/A_{i,t-1}] + \gamma_6\left(\frac{gppe_{i,t}}{A_{i,t-1}}\right) + \mu_{i,t}. \quad (C.5)$
CAPX	Capital expenditure scaled by lagged total book assets, where capital expenditure is measured by change in net property, plant, and equipment plus depreciation. (Source: Compustat)
$\Delta CAPX$	Change in capital expenditure, scaled by lagged total book assets. (Source: Compustat)
Cash	Cash and marketable securities, scaled by lagged total book assets. (Source: Compustat)
$\Delta Cash$	Change in cash and marketable securities, scaled by lagged total book assets. (Source: Compustat)
Fscore	The F-score calculated based on model 1 of Dechow et al. (2011). Specifically, for each firm-quarter, we use model 1 in table 7 of Dechow et al. (2011) to calculate the predicted value z of the model as follows:
	$z = -0.7893 + 0.790 \times RsstAcc + 2.518 \times ChRec + 1.191 \times ChInv + 1.979 \times SoftAssets + 0.171 \times ChCs - 0.932 \times ChRoa + 1.029 \times Issue, \quad (C.6)$
	where $RsstAcc$ denotes the accruals calculated based on Richardson et al. (2005), $ChRec$ is the change in receivables during the period (from the end of quarter $t-1$ to the end of quarter t) scaled by average total assets during the period, $ChInv$ is the change in inventory during the quarter scaled by average total assets during the period, $ChCs$ is the percentage change in cash sales during the quarter, $ChRoa$ is change in return on assets from quarter $t-1$ to quarter t , $SoftAssets$ is calculated as (total assets – gross property, plant, and equipment – cash and cash equivalent) scaled by total assets at the end of the quarter, and $Issue$ is an indicator variable coded 1 if the firm issued securities during the quarter. Because the model is estimated based on a logistic regression, the probability of misstatement Y is the cumulative standard logistic distribution function, evaluated at predicted value z . That is, $Y = 1/(1 + e^{-z}) = (e^z)/(1 + e^z)$. We then use the unconditional probability 0.0037 provided by Dechow et al. (2011) to calculate the F-score as $Y/0.0037$. (Sources: Compustat and Dechow et al. 2011)
HighF_20p	A dummy variable that takes the value of 1 if the firm's Fscore ranks among top 20% of the sample in the quarter.
$r - R^{IND}$	A firm's quarterly stock return minus the corresponding value-weighted average industry return in the same quarter, where industries are defined based on the Fama–French 48-industry classification. ^a
$r - R^{Size,BM}$	A firm's quarterly stock return minus the corresponding value-weighted Fama–French size and book-to-market portfolio return in the same quarter. ^a
Net financing	Total equity issuance minus repurchases plus debt issuance minus debt redemption, scaled by lagged total book assets. (Source: Compustat)
$\Delta Earnings$	Change in earnings before extraordinary items plus interest, deferred tax credits, and investment tax credits, scaled by lagged total book assets. (Source: Compustat)
$\Delta NetAsset$	Change in net assets, scaled by lagged total book assets. Net assets are total assets minus cash holdings. (Source: Compustat)
$\Delta R\&D$	Change in R&D expenditure, scaled by lagged total book assets. (Source: Compustat)
$\Delta Interest$	Change in interest expense, scaled by lagged total book assets. (Source: Compustat)
$\Delta Dividend$	Change in common dividends paid, scaled by lagged total book assets. (Source: Compustat)
Interim	For firms that filed in EDGAR before 1994, this dummy variable takes value 1 from Q3 1993 to Q1 1994 and 0 otherwise. For firms that began filing in EDGAR after 1994, this variable takes value 0.
Monthly frequency	
logAmihud	Natural logarithm of the Amihud illiquidity measure. For each stock, we first calculate the ratio of its daily absolute return to its dollar trading volume on the day. We then average the ratio over the month, multiply it by 10^6 , and apply a natural log transformation. (Source: CRSP)
Effective bid-ask spread (%)	Relative effective spread (in percent). It is calculated as twice the difference between the execution price and the midpoint of the prevailing best quotes divided by the midpoint of the prevailing best quotes, averaged over all intraday spreads for the day and then over all trading days in the month. (Source: TAQ)
Quoted bid-ask spread (%)	Relative quoted spread (in percent). It is calculated as the spread of the national best bid and ask quotes divided by the midpoint of the quotes, averaged over all intraday spreads for the day and then over all trading days in the month. (Source: TAQ)
Ret	Stock returns during the month. (Source: CRSP)
Prc	Stock price at the end of the month. (Source: CRSP)
StdRet	Annualized stock return volatility. It is estimated by the standard deviation of the daily stock returns during the month multiplied by the square root of 252. If there are fewer than 15 nonmissing daily returns in a given month, we set it as a missing value. (Source: CRSP)

Notes. This table provides the definitions of the variables used in our analyses. Accounting variables are defined similarly for quarterly and monthly data and are measured based on the latest data available from Compustat prior to the end of the period.

^aSee Professor Kenneth French's website available at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html (accessed September 2021).

Endnotes

¹ The EDGAR system took almost 10 years to develop and was a major effort by the SEC to facilitate investor access to corporate financial information (SEC 1993).

² Frydman and Saks (2010) document that stock and stock option grants have represented a large part of executive compensation since 1980. The value of these holdings is linked directly to stock prices.

³ As discussed in more detail in Section 3.1, there were two incidences of nonrandom assignment: firms participating in EDGAR's pilot program were assigned to the first group, and smaller firms were assigned to the last group to give them more time to adjust to the new filing method. We exclude these two groups of firms from our empirical analysis.

⁴ Our findings suggest that firms' investment efficiency increases during the post-EDGAR period. It is noted that although Goldstein et al. (2022) document a decrease in the post-EDGAR investment Q sensitivity, they show that it was a result of stock prices containing less new information to managers during the period.

⁵ Blankespoor et al. (2014) and Li et al. (2012) differ in their sample construction and empirical methodologies. The former exclude voluntary adopters from their sample out of concerns of self-selection bias and employ a DiD test methodology. The latter include voluntary adopters in their sample and conduct all tests using pooled regression models.

⁶ Liu et al. (2014) show that the cost of capital increases after XBRL adoption, but Chen et al. (2015) find the opposite.

⁷ Given that stock analysts provide a wider and more accurate coverage of listed firms and that stock prices are more informative post-EDGAR, disclosure integration (i.e., analysis) costs are likely to decline for both small and large investors.

⁸ For example, Mead Data Central's Nexis/Lexis data service and SEC Online offered full-text access to some corporate filings. The former charged a monthly subscription fee of \$125, plus a connection charge of \$32 an hour, plus \$14 per filing, plus 2 cents per line of data for viewing or printing. The latter could be accessed via the DIALOG data service, which charged a monthly subscription fee plus a flat fee of \$120 per hour of connection time. Moreover, subscriptions to abstracted information of some SEC filings were available from some data vendors. Investors could access these data by paying a monthly subscription fee plus an online connection fee of approximately \$45–\$126 an hour, plus \$0.78–\$17 per record printed offline or \$0.68–\$13 per record displayed online. In 1992, Compustat offered income and balance sheet data of selected firms on CD-ROMs and data on annual reports, 10-K, 10-Q, and proxy statements at \$9,000–\$18,000 per year (<http://www.bio.net/bionet/mm/ag-forst/1992-January/000187.html>).

⁹ One of the key assumptions in theorem 1 of Athey and Imbens (2022) is that treatment effects are constant over time. In Section 7.3 and Table 12, we carry out additional tests based on Cengiz et al. (2019) and Callaway and Sant'Anna (2021), which are robust under dynamic treatment effects.

¹⁰ Because firm risk may be affected by EDGAR implementation, we set the control variable $StdRet_{i,t-1}$ at its latest value prior to quarter k , where $k = \min(t, \text{event year-quarter } t^*)$.

¹¹ We have 1,802 matched pairs of treated and control firms in this sample. The list of matching characteristics is the same as that reported in panel A of Table 3 except that $FBias$ is replaced by Prc . The (untabulated) balance test results show that the characteristics are not statistically different between treatment and control firms in this sample.

¹² We define the variable $StdRet_{i,m-1}$ as its latest value prior to month k , where $k = \min(m, \text{event year-month } m^*)$.

¹³ We end the sample period of the first and second falsification tests in Q1 1993 and start the third falsification test in Q3 1996 to avoid overlapping with the actual event period. We start the second falsification test in Q1 1989 to avoid unnecessary complications from the stock market crash of 1987–1988. There are relatively few observations in the third falsification test because we require (i) that a treated firm have at least one quarter of data before and after the pseudo-EDGAR event, (ii) that a control firm has not changed its filing method four quarters prior to the event quarter and four quarters afterward, and (iii) that all observations after Q4 1998 be dropped to avoid any estimation bias that might arise as a result of the internet bubble of 1999–2000.

¹⁴ Because a firm i can only be a treated or a control firm but not both in this specification, the treated firm dummy $Treat_i$ needs to be removed from Equation (3) to avoid perfect multicollinearity.

¹⁵ We obtain the EDGAR index file from the SEC's website at <https://www.sec.gov/os/accessing-edgar-data>, which provides the company name, form type, and date filed of each public filing available in EDGAR from 1993 onward.

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